Allianz Trade

Allianz Trade Global Survey 2026

Business as unusual

How exporters adapt to geopolitical shocks

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Allianz Research

Content

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Executive Summary


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Early signs of disruption. The Middle East conflict has added a new layer of shocks to an already fragile environment shaped by tariffs, weakening demand and declining consumer confidence. We expect lower global GDP growth (+2.6% in 2026), higher global inflation (4.3% in 2026) and stronger fiscal pressure, with higher energy and input costs and weak demand adding to the pressure of 10.5% effective US tariffs on companies' margins. Even in the best-case scenario, a post-ceasefire recovery in the Strait of Hormuz would take time (reaching 15-30% of normal levels). Against this backdrop, for the 5th edition of the Allianz Trade Global Survey, we asked 6,000 companies in Brazil, China, France, Germany, India, Italy, Poland, Singapore, Spain, the UAE, the UK, the US and Vietnam about their outlook for 2026, before and after the outbreak of war.

- **Export confidence** has held up better than during the 2025 tariff shock - dropping only 6pps to 75% of exporters still expecting positive growth - compared to the 40pps collapse after "Liberation Day." However, the impact is uneven: Vietnamese, American and Spanish firms lost more than 10pps of confidence, while Chinese firms, already weakened by the trade war, lost 9pps to 51%.
- **Logistics and energy** are the most immediate concerns. 60% of firms are worried about supply-chain disruption and rising energy and commodity prices. In the wake of the war Iran, countries are faced with different challenges. Some are highly exposed and with low buffers (e.g. Vietnam, Thailand etc.), others are exposed but have buffers through reserves, alternative suppliers etc. (e.g. European countries, China etc.). Against this backdrop, Vietnamese (79%), Polish (76%), British (72%) and American (71%) firms show high levels of concern. In contrast, Indian and Chinese firms appear relatively less worried.
- **Operational adjustments have accelerated.** Over half of companies are now seeking alternative shipping routes or carriers – especially in Vietnam (60%), the US and India (55% each). Many are also working with customs brokers to expedite clearance (Vietnam 64%, India 56%) or adjusting delivery schedules. These operational responses are moving faster than contractual changes.
- **Trade finance conditions are tightening.** The share of firms expecting payment terms to deteriorate has rebounded to 43% (+5pps since the conflict began), with the sharpest rises in Brazil (+18pps), the UAE (+10pps), India and Vietnam (+9pps each). Non-payment risk fears have risen to 40% of firms (+6pps vs. pre-conflict), with the most exposed sectors being pharmaceuticals, construction, and computers/telecoms.

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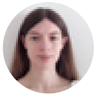

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- **Reshoring dynamics have shifted.** The conflict has accelerated reshoring intent, particularly in Europe – Poland, the UK and France lead this shift – while US and Vietnamese firms moved in the opposite direction. The UAE shows a bifurcated response, reflecting its dual role as both a logistics hub and a geography directly exposed to the crisis.
- **AI optimism has taken a hit.** The share of firms expecting AI to drive export growth of +10% fell 8pps post-conflict, from ~30% before the war.

Beyond the conflict, global trade has changed for good. We identify seven lessons from this year's survey:

1. Risk landscape: Geopolitics now dominates

The risk hierarchy has been reshuffled since 2025. Geopolitical and political risk - wars, tariffs, expropriation, social unrest – now tops the list for 65% of firms (+11pps), displacing supply-chain complexity, which fell to third place (45%, -30pps). Supply-related risks – supplier bankruptcies and input shortages – surged to second place (57%, +30pps), consistent with record-high global insolvencies running 24% above their pre-pandemic average. The economic cost of supply-chain complexity reached USD4.7trn in 2025, more than double its 2017 level, with 56% linked to US trade flows.

2. One year of US tariffs: Supply chains are shifting...

80% of firms have adjusted their trade and supply-chain routes since "Liberation Day" to avoid higher tariffs and geopolitical risk. Despite a Supreme Court ruling against IEEPA tariffs, Section 122 tariffs have kept the US rate stable at ~9% until at least end-July 2026. 43% of firms still expect a net negative impact from the trade war - higher than the 39% before the trade war began. Concerns are sharpest in China and Germany (50% and 49% respectively). Nevertheless, fewer firms (32%, -6pps) plan to raise prices due to tariffs, back to pre-"Liberation Day" levels, while companies appear more growth-oriented in their investment strategies: capex priorities have risen (28% vs. 20% in 2025) while cost-cutting has declined (26% vs. 31%).

3. ...and de-risking has become the norm

Seven out of ten firms have taken operational steps to adapt since the trade war began. The most common strategies remain inventory building and market diversification (64%), sourcing from new suppliers (63%) and rerouting through third markets (57%). China leads in rerouting (69%) and market diversification (75%), while Germany has the lowest diversification rate (52%). On Incoterms, DDP usage by exporters has dropped sharply from 25% to 16%, reflecting reluctance to absorb tariff liability, while FOB adoption by importers has grown (30%, +10pps) as buyers seek greater control over logistics.

4. Geographic reorientation: US loses ground, Europe and Asia gain

The US did not gain back any appeal: only 13% of firms now consider it a growth market (vs. 17% in 2025). Interest in Europe has grown, led by Singaporean (+10pps) and US (+9pps) exporters. Asia-Pacific excluding China is the clearest structural beneficiary of supply-chain realignment - Vietnam, India, Indonesia and Malaysia are gaining investment flows. China's appeal has collapsed: self-retention among China-based firms fell from 32% to 16%, while only 23% of firms globally plan to increase their footprint there (-30pps from 2025). A new wave of FTAs - India-EU, MERCOSUR-EU - is capturing attention, with 93% of firms planning to use them to expand, though non-tariff barriers (licensing, certification) remain the dominant friction.

5. Payment terms and financial risk: Structural lengthening

Payment cycles are lengthening structurally. Only 7% of companies are now paid within 30 days (-4pps vs. 2025), while nearly one in four (24%, +7pps) are paid after 70 days. Larger firms are disproportionately affected: 42% of companies with turnover above EUR3bn face payment terms exceeding 70 days. The most exposed sectors for long payment delays are transport equipment, pharmaceuticals, and computers/telecoms. Bank loans (46%) and internal cash flows (44%) remain the dominant financing sources, while state support has declined in importance since last year's survey.

6. ESG: A fractured consensus

After years of convergence, the global ESG consensus has broken down. Overall ESG commitment fell 22pps to 62% (from 84% in 2025), with the sharpest drops in China (-42pps to 47%) and the UK (-29pps to 55%). European firms are holding firmer - Germany fell only 9pps to 76%. The divergence is driven by the US stepping back from federal sustainability frameworks while the EU advances, creating multiple incompatible regulatory regimes. Firms are prioritizing supply-chain measures (59%) over deeper internal reforms – governance (34%) and executive incentives (29%) lag. Despite this, climate ambition remains intact: 26% of firms target CO_2 reductions of 5–10% (+4pps), and 84% remain confident of reaching net zero.

7. AI: A two-speed adoption

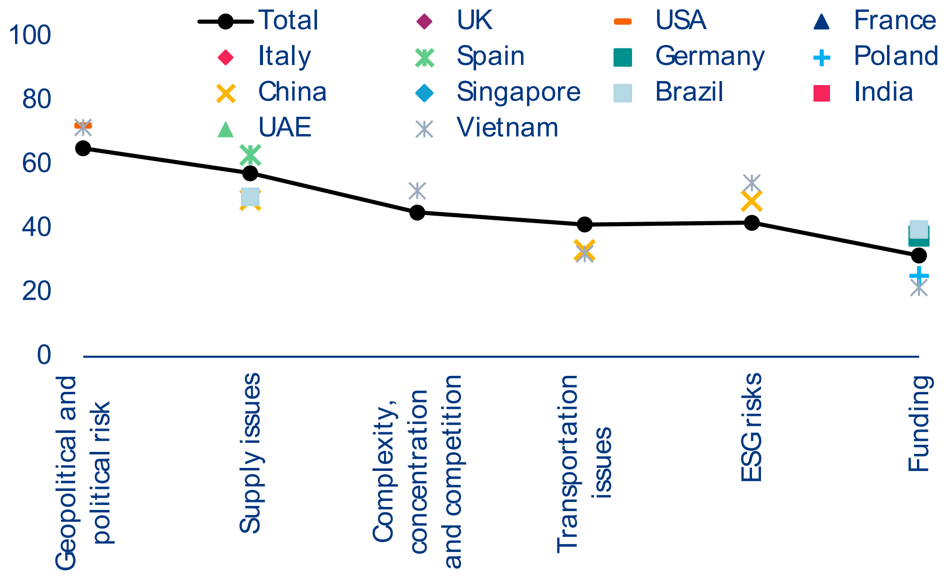
AI adoption is now near-universal - only 0.5% of exporters report not using it - but depth and strategic intent vary sharply. Emerging economies lead: UAE (86%), Poland (80%) and India (75%) report the highest scaled deployment rates, while the UK (57%) and US (63%) remain below 65%. The real divide is expectations: 61% of Indian firms anticipate AI will boost export turnover by 10%+, against just 18–22% in Europe. High adoption does not always translate to growth optimism - UAE firms deploy AI at scale but remain cautious on impact (22%), suggesting efficiency-focused rather than growth-oriented use. The universal barrier is not cost or skills but ROI uncertainty, cited by 28% of firms globally.

A new layer of shocks in an already fragile environment

War and military conflicts were named the larger threat to companies' export activity and supply chains over the next two years (35%) than the trade war and protectionism (28%). For the 5th edition of the Allianz Trade Global Survey, we asked 6,000 companies in Brazil, China, France, Germany, India, Italy, Poland, Singapore, Spain, the UAE, the UK, the US and Vietnam about their outlook for 2026. Since the start of the conflict, 60% of firms have raised concerns surrounding the logistics consequences of the war for supply chains and the rise of energy and commodity prices. In the wake of the war Iran, countries are faced with different challenges. Some are highly exposed and with low buffers (e.g. Vietnam, Thailand etc.), others are exposed but have buffers through reserves, alternative suppliers etc. (e.g. European countries, China). Against this backdrop, our survey reveal that Vietnam and Poland are the most worried closely followed by firms in the UAE (68%). Furthermore, British (71%) and American (72%) firms also appear particularly worried by the aftermath of the conflict. In contrast, Indian (36%) and Chinese (39%) firms appear relatively less worried. While Indian and Chinese imports are largely dependent on the Middle East region (nearly 50% for each), the reliance on coal in their energy mix and their ability to negotiate with Iran have allowed them to remain slightly more shielded from the aftermath of the closure of Hormuz than other countries of the sample.

Geopolitical and political risks have emerged as the largest threat to offshore production and supply-chain stability, cited by 65% of surveyed firms and now ranking as the leading concern overall (figure 1). Compared to the 2025 edition of the survey, geopolitical and political risks, such as conflicts and tariffs but also internal vulnerabilities such as expropriation and social unrest, rose by +11pps, displacing supply-chain complexity and concentration as the top-ranked threat, with the latter falling by -30pps to 45%. Simultaneously, supply-related risks, such as supplier bankruptcies and input shortages, climbed to second place, identified by 57% of firms and up +30pps from last year. This rise is consistent with our own analysis of global business insolvencies, which have reached record highs, running 24% above their pre-pandemic average. Following a +10% increase in 2024, insolvency levels remained elevated with a further +6% rise in 2025. Looking ahead, we expect this pressure to persist throughout 2026, with our baseline forecasting a +5% increase, +2pps since the escalation of the Middle East, before stabilizing in 2027.

Figure 1: Geopolitical and political risks have become the largest threats to companies' business



Sources: Allianz Trade Global Survey 2026

Notes: Geopolitical and political risks refer to war, tariffs, confiscation, expropriation and social unrest. Supply issues refer to bankruptcy of suppliers and shortage of inputs. Only countries deviating by more than 5pps compared to the sample average are shown.

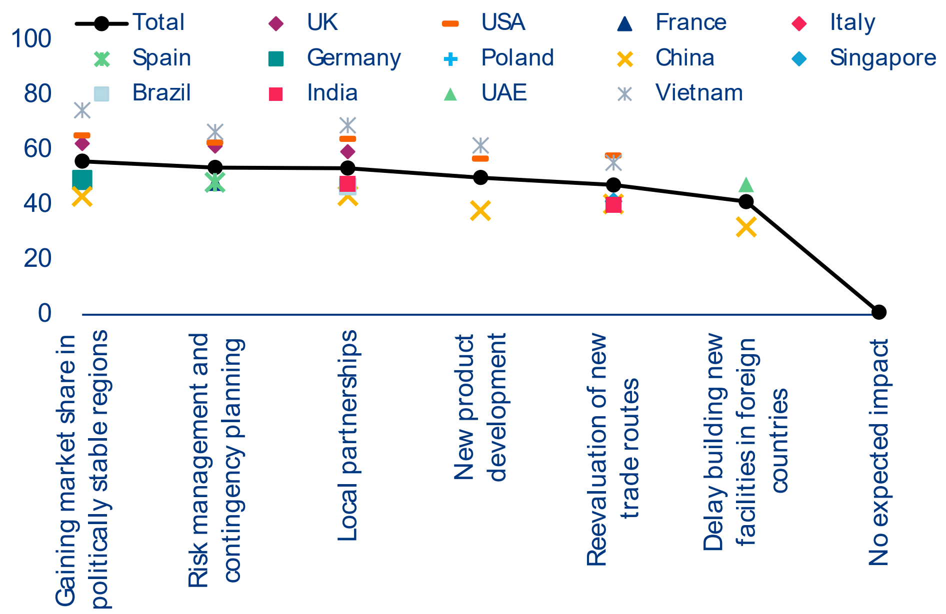
We estimate that the economic cost of supply-chain complexity has surged to USD4.7trn in 2025, more than twice its 2017 level, with 56% attributable to trade flows linked to the US. Our proprietary index indicates that global supply-chain complexity continued to intensify in 2025, more than doubling compared to 2017 and reaching its highest level since 2014. The index captures weighted bilateral trade flows, adjusted for geographical and geopolitical distance, as well as infrastructure quality and connectivity, climate vulnerability, growth prospects and trade openness.

At the country level, geopolitical and political risks weigh most heavily on firms in the US and Vietnam (for 72% of firms for each country). Brazilian firms sit modestly below the global average at 60%. Worries around supply disruption are most acute in Spain (63%), while firms in China (49%) and Brazil (50%) appear comparatively less exposed on this dimension. On complexity, concentration and competitive pressures, firms in Vietnam register the highest level of concern in the sample at 52%.

Nevertheless, companies still expect positive turnover generated through exports, though Chinese firms are more downbeat. Before the escalation in the Middle East, 80% expected an increase of at least +2% in turnover generated through exports. But since the crisis broke out, firms' expectations have only fallen by -5pps to 75%. This is a much brighter outlook than right after "Liberation Day", when only 40% of firms expected an increase. Across countries, the results are broadly similar, with expectations of turnover growth above +2% generally ranging from 70% to 85%, and expectations of lower turnover ranging from 3% to 10%. Indian and German companies are the most upbeat (82-83%). China and Vietnam stand out as the show a significant deterioration of turnover expectations since the start of the war in Iran. In China, only 51% of firms surveyed have an upbeat outlook on their turnover generated by exports (turnover growth > 2%), significantly lower than just before the Middle East conflict (-9pps), though improving from post "Liberation Day". Furthermore, 8% of Chinese firms expect negative turnover – similar to their Brazilian counterparts (9%) – though less than Vietnamese companies (13%). Beyond the impact of the war in the Middle East, one explanation for the pessimism of Chinese companies could be the widespread strategy of aggressive price cuts, notably amid elevated US tariffs.

Geopolitical threats are prompting firms to focus on their local markets, stay closer to home or prioritise gaining market share in politically stable regions (56% of respondents). Virtually all firms are adapting to the ongoing geopolitical risks, using all tools in hand from risk management to more active strategies. Enhanced risk management and contingency planning (54%) is among the preferred options, especially by Chinese firms. Meanwhile, other more active strategies including new product development and new markets and routes are being considered by at least half of the surveyed firms. Reconsideration of investments and expansion is the least preferred option, though 40% of firms are still considering this.

Figure 2: Companies' priorities to operate and grow internationally despite geopolitical risks



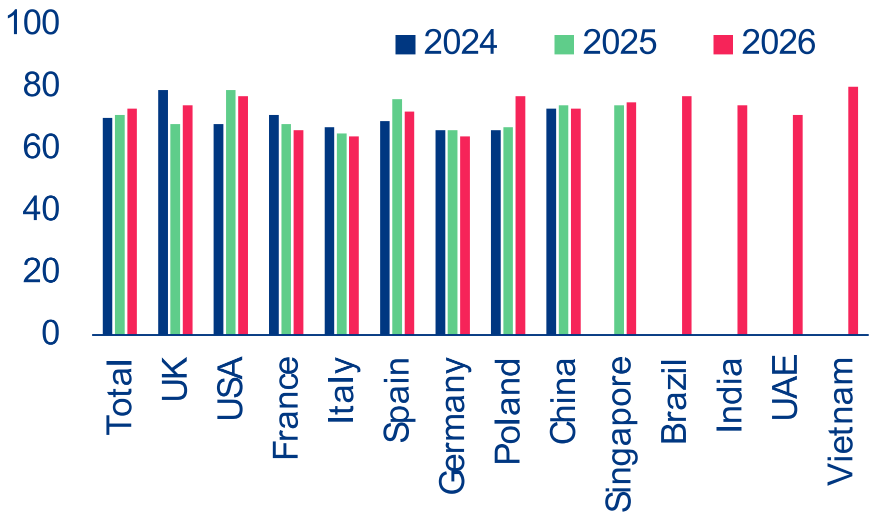
Sources: Allianz Trade Global Survey 2026
Notes: Only countries deviating by more than 5pps compared to the sample average are shown.

Since the start of the conflict, firms have intensified operational adjustments, particularly in logistics and trade execution, however patterns vary across different geographies. More than half of companies are seeking alternative shipping routes or carriers especially in Vietnam (60%), the US (55%), India (55%) and Brazil (52%), while many are working with customs brokers to expedite clearance or adjusting delivery schedules especially in India (56%) and France (53%), indicating that operational responses are accelerating faster than contractual changes.

Vietnam stands out as the most proactive economy, showing among the highest readings across virtually every strategic priority, with a pronounced preference for capturing market share in politically stable regions (74%). The only exception to this is the decision to delay or reconsider foreign facility construction, where Vietnam records the second-lowest score (38%). At the other end of the spectrum, Chinese firms display a more cautious posture, consistently registering the lowest or near-lowest response rates across categories. Their most frequently cited priority, namely enhanced risk management and contingency planning (49%), points to a defensive rather than opportunistic strategic orientation in the face of geopolitical disruption. Similarly, the UAE (47%), Italy (46%), Poland (46%) and the US (45%) emerge as the most inclined to pause capital commitments. Yet the US picture is more nuanced with firms simultaneously ranking as the second most likely to accelerate new product development (57%), behind Vietnam (62%), and leading the sample in reevaluating trade routes (58%), ahead of Vietnam (55%) and the UAE (52%).

While reshoring dynamics have remained strong, with 72% confirming, the Middle East crisis has caused the sharpest increase towards reshoring acceleration, especially in Europe, where Poland, UK, and France lead the reshoring boost. China and Singapore also pointed towards an acceleration of reshoring caused by the war. While the US and Vietnam moved in the opposite direction. Vietnamese firms had the highest pre-war acceleration expectations because they are the primary China-decoupling beneficiary, but the Middle East conflict complicated their position by introducing geopolitical risk. The UAE shows an unusual bifurcation: acceleration and reverse of the trends jumped reflecting the country's contradictory position as both a logistics hub benefiting from supply chain restructuring and a geography directly exposed to the Middle East crisis.

Figure 3: Share of firm expecting the reshoring trend to accelerate in the coming years



Sources: Allianz Trade Global Survey 2026

However, structural constraints continue to limit the pivot towards domestic suppliers and full reshoring. Most pressing issues across all geographies continue to be limited access to competitive domestic suppliers (around 83%). Production costs and tax incentives form the second tier of constraints. The UK and Brazil record the highest production cost concern, while India and Poland highest tax incentive concern. A notable shift caused by the Middle East crisis has been US firms concern about ESG, with a drop in production cost concerns as Middle Eastern supply-chain disruption risks making offshore sourcing more expensive and reputationally complex, enhancing the compliance and reputational risks of supply-chain exposure.

Paying the price in payment terms

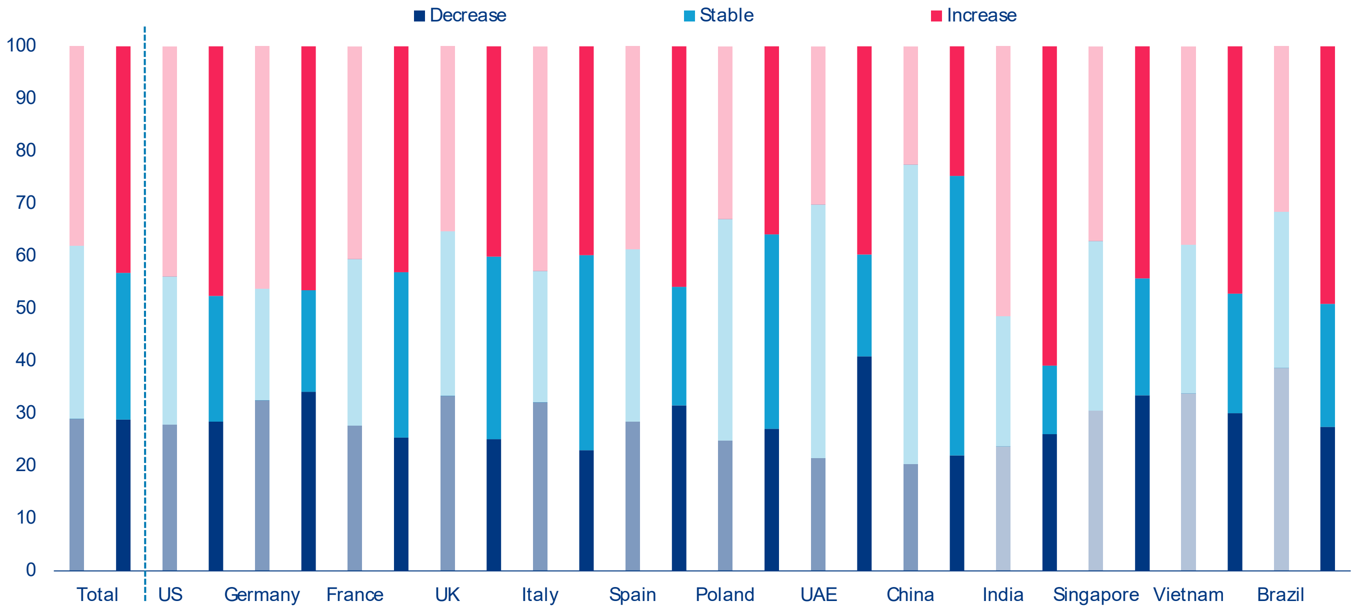
The war in the Middle East has reversed expectations in payment terms among exporters but more moderately and unevenly than after the start of the trade war last year. Prior to the start of the war in the Middle East, just 38% of respondents were expecting export terms to increase in the next 6 to 12 months and 31% said they would remain stable, an improvement compared to the last year survey after "Liberation Day" (53% and 23% in 2025, respectively) and almost a return to the pre-"Liberation Day" sentiment (37% and 31%, respectively). Germany was the one exception, with a deterioration compared to both references. However, the war has changed the picture, with the overall share of exporters worried that the length of payments will increase rebounding to +43% (+5pps), though this is still below the 53% recorded last year after "Liberation Day". This rebound comes from mixed dynamics, with (i) the largest rises in Spain, Singapore, India, Vietnam, the UAE and Brazil (+7, +7, +9,+9, +10 and +18, respectively); (ii) noticeable increases in China, France, the US and the UK (+2, +3, +4 and +5, respectively) and (iii) roughly unchanged expectations in Poland and Germany.

Overall, companies are more pessimistic than last year (41% versus 36%), notably in Western Europe. In most countries, exporters are mostly fearing a rise in payment terms. This is in particularly the case in India (56%), Germany (46%), the US (46%), Spain (44%) and France (42%). In Poland and China, most firms are expecting a stable outlook in payment terms (40% and 55%, respectively). All in all, exporters anticipating a decrease in payment terms never exceed one-third of the panel, with the highest share in Vietnam and Germany, and the lowest share in France and India. This deterioration in payment terms is widespread across all company sizes, especially those with fewer than 250 employees: 45% of them expect export payment terms to rise in the next six to twelve months, compared to 70% for the firms with 250+ employees. It is also widespread across sectors, with a majority (11 out of 15) mostly bracing for an extension in payment terms, notably in paper, construction, pharmaceuticals, agrifood and computers/telecom (61%, 59%, 51%, 50% and 47%, respectively). In addition, this extension in payment delays is expected to be significant, exceeding seven days in almost half of the cases, notably for computer telecom (55%), pharmaceuticals (52%) and transport equipment (50%). At the same time, a larger share of respondents in the automotive and machinery equipment sectors expect the length of payment terms to remain stable (47% and 39%, respectively). Three sectors stand out with a relatively more equal distribution of expectations in the length of payment terms (increase/stable/decrease): metals, household equipment and trade. Interestingly, energy and transport equipment both post great divide among their firms, with the same and a large proportion of companies expecting either an increase or a decrease (roughly 40%).

The perception of export non-payment risk has also deteriorated with the war in the Middle East, with fears of higher non-payment risks up by +6pps – despite varying reactions across sectors. 40% of companies are now anticipating more non-payment risk over the next six to 12 months – compared to 34% prior to the start of the war in the Middle East and 38% in 2025 prior to "Liberation Day". The deterioration is most visible in Brazil (higher non-payment risk: from +33% to 54%) and the UAE (from 23% to 43%), followed by India, Vietnam and China, while the US and European countries have shown relatively more stability since the start of the war. Both Brazil and the UAE experienced a regime switch, with expectations in stable non-payment risk collapsing (from 54% to 38%, and from 69% to 44%, respectively). India, the US and Brazil now record the highest share of firms expecting a rise in non-payment risk (54%, 47% and 45%, respectively) while France and Germany stand out with higher numbers than post "Liberation Day" (41% and 40%, from 34% and 37%, respectively). Overall, a majority of firms still expect the risk of export non-payment to remain stable in the next six to twelve months (52% versus 56% prior to the start of the war in the Middle East).

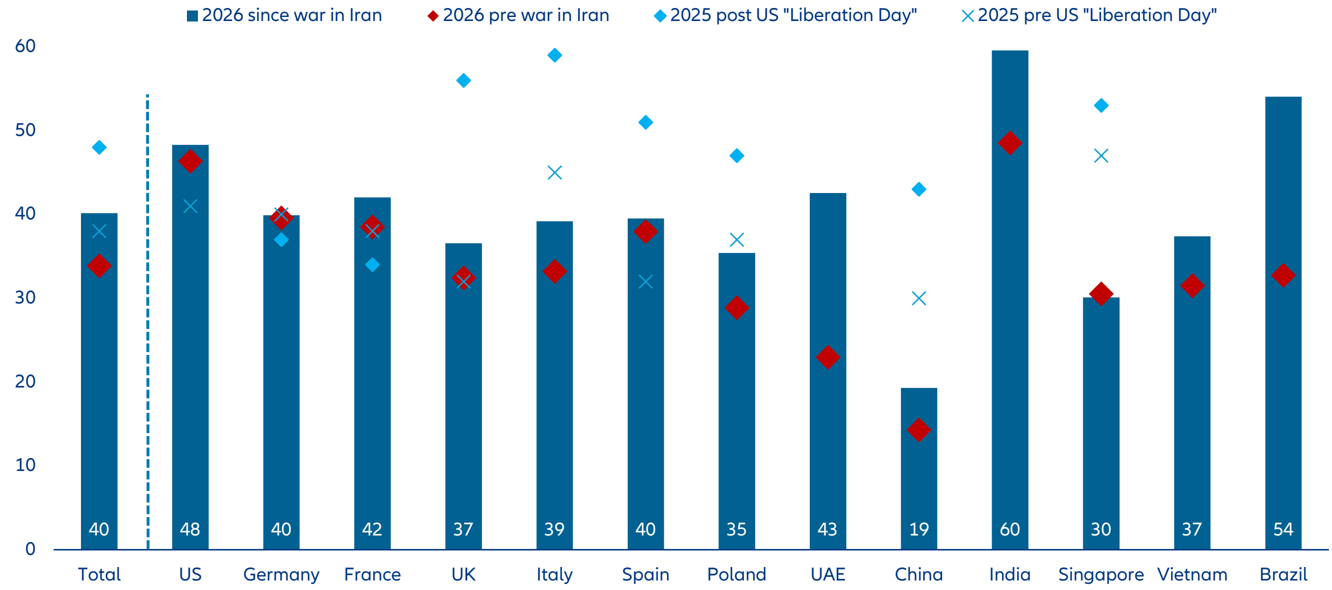
Meanwhile, companies in Italy, the UK, Singapore and Poland (55%, 57%, 60% and 61%, respectively) but also, and above all, in China (73%) expect a stable risk of non-payment. The flip side of this picture is that in all countries a minority of firms, most often below 10%, expect a decrease in non-payment risk. Interestingly, exporters in pharmaceuticals (55%), construction (52%), computers/telecom (44%) and transport equipment (44%) are more worried about the risk of non-payment increasing than exporters in machine equipment (31%), trade (33%), household equipment (33%) and textiles (34%). The war in the Middle East has however contributed to deteriorate expectations (i.e. higher risks) in a majority of sectors, in particular energy (+22pps to 42%), household equipment (+13pps to 33%), computers/telecoms (+12pps to 44%), and machinery equipment (+10pps to 31%). Chemicals, construction, transport equipment and pharmaceuticals are the exceptions. All sectors combined, the larger (lower) the turnover, the higher the share of greater (lower) non-payment risk – but the gap has reduced noticeably since the start of the war in the Middle East.

Figure 4: Expectations of change regarding the length of export payment terms in the next six to 12 months, % of total companies, before and after the start of the war in the Middle East



Sources: Allianz Trade Global Survey 2026

Figure 5: Share of respondents expecting the risk of export non-payments risk to rise in the next six to 12 months, % of total companies, 2026 vs 2025



Sources: Allianz Trade Global Surveys 2025 and 2026

Only 7% of companies continue to be paid within 30 days. The share is even lower in Italy and Singapore (4% and 5%, respectively), and to a lesser extent Brazil and Germany (6% each), but Vietnam and the UK report a higher share (10% and 11%). This outcome represents a decrease compared to last year's survey (-4pps), with strong decreases for Italy (-8pps from 12% to 4%) and Singapore (-10pps from 15%), as well as declines in the US (-1pp), China (-2pps), Germany (-3pps), France (-4pps), Poland (-4pps) and Spain (-5pps) – and the UK as main exception (+1pp).

Roughly 70% of companies are paid between 30 and 70 days (68%), with the share slightly higher in Poland (76%), the UK (75%) and the US (75%), while China and India stand out on the opposite side (59% and 46%, respectively). This second outcome also results from a drop compared to the previous survey, for our extended panel (-3pps) as well as for most countries such as China, Singapore, Germany and France (-4, -2, -2 and -2, respectively) – despite some exceptions for Spain, the US, Poland and Italy (+1, +2,+3 and +4, respectively).

At the opposite end of the spectrum, longer payments have gained traction. Almost one out of four export companies are paid after 70 days (+7pp to 24%), with a better situation for firms in advanced economies (20%) than in emerging markets (30%). But the latter displays large discrepancies from India to Vietnam (46% and 22%, respectively). The longest export payment delays increased in most countries, compared to 2025, with 5% of companies paid after 90 days (i.e. +1pp), notably in India, Brazil and Spain (16%, 8% and 6%, respectively) – and noticeable rises in Germany, France and Spain (+3pps each to 4%, 5% and 6%, respectively). Firms in transport equipment, pharmaceuticals, computers and telecom, household equipment and automotive often wait the longest to be paid, with a noticeable share of them paid after 90 days (23%, 17%, 8% and 7%, respectively).

Overall, roughly 70% of companies are paid between 30 and 70 days, with a stronger concentration for firms in the paper, textiles and construction sectors (89%, 81%, and 78%, respectively). Firms in the agrifood, retail and computers and telecom sectors see shorter export payment terms, with the highest share of respondents reporting less than 30 days (9%, 8%, and 8%). All sectors combined, the larger (lower) the turnover, the higher the share of longer (lower) export payment terms, with 42% of surveyed companies having a turnover above EUR3bn facing payment terms exceeding 70 days. This hides a wide dispersion, with a higher share of firms in this situation in transport equipment (83%), machinery equipment (51%), automotive (55%), construction (53%). 54% of surveyed companies with a turnover above EUR5bn face payment terms exceeding 70 days, compared to 25% for the overall sample average and 21% for the companies with less than EUR3bn in turnover.

Bank loans (46%) and internal cash flows (44%) remain the dominant sources of financing for companies. Reliance on bank lending is particularly strong in China (56%) and Vietnam (53%), consistent with their bank-centric financial systems and the prominent role of state-backed credit. In contrast, internal financing is more prevalent among UK, Singaporean (51%), US and Polish (49%) firms, reflecting deeper capital markets and stronger corporate balance sheets. State support is most frequently cited by UAE (38%) and Singaporean (41%) companies, highlighting a significant government role in supporting economic activity. Compared to last year's survey, the overall ranking of financing sources remains broadly unchanged at the top, with bank lending and cash flows still leading, while state support has declined from the third to the sixth most cited option. More sophisticated financing solutions, such as private equity, are used by 35% of firms and are particularly popular in emerging economies, including Vietnam (43%), India (44%), and Brazil (38%).

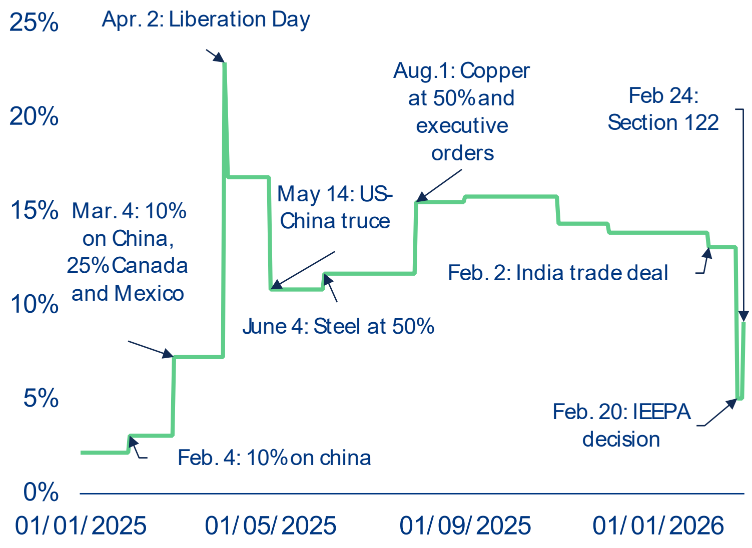
In terms of foreign exchange risk management, improving forecasting and internal risk management processes remains the primary strategy (58%), followed by FX hedging (48%) and the use of common currencies for payments and receipts (48%). These preferences have remained broadly stable since the onset of the conflict in the Middle East, suggesting limited short-term adjustment in corporate FX practices. Cross-country differences nevertheless stand out. Indian firms show a stronger preference for using local currencies in transactions, likely reflecting persistent depreciation pressures on the rupee. At the other end of the spectrum, Chinese (36%) and UK (33%) firms make less use of local currency payments. Despite increasing discussions on the internationalization of the renminbi, Chinese exporters continue to rely heavily on the USD. This is reflected in the relatively low share of Chinese firms increasing non-dollar transactions (28%, compared to a 37% sample average).

The consequences of a year of US tariffs

Over the past year, US trade policy has undergone a large shift, defined by the tariff offensive initiated on "Liberation Day". At its peak, the US applied a global tariff rate of 23%. Subsequent developments have brought meaningful, if partial, relief: we estimate the US tariff rate now stands at 9%, following the Supreme Court ruling in February 2026 and the imposition of Section 122 tariffs in its wake

The impacts of US tariffs continue to reshape global trade and supply-chain routes, with 80% of firms having introduced new routes to avoid higher tariffs, geopolitical risks and/or improve capacity. Firms in Vietnam (88%) and India (87%) have been the most reactive, followed closely by French and American exporters (85%). 68% of Chinese exporters have adjusted routes, though they were already doing so since the first Trump administration in 2017. In addition, Chinese exporters that did not yet make significant changes to their existing supply chains and trade routes are the most numerous across the sample to consider doing so in 2026 (16%, above the global average of 10%).

Figure 6: Evolution of the global tariff rate applied by the US



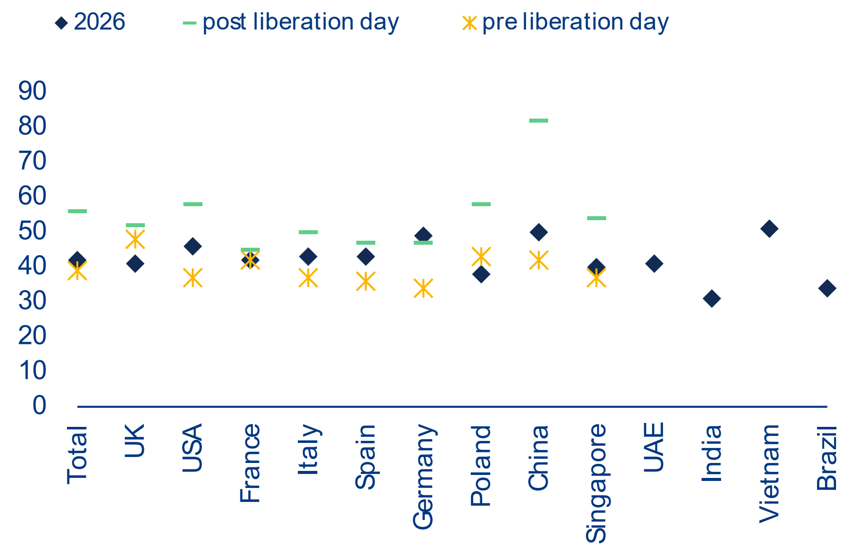
Sources: Allianz Reserach

43% of firms anticipate a net negative impact on their business, still higher than the 39% before the start of the trade war in 2025. Despite the recent Supreme Court decision to declare IEEPA tariffs illegal, Section 122 has reimposed tariffs on countries across the board for 150 days (until end-July 2026), which keeps the US average tariff stable at 9%. Concerns are most acute in China and Germany, where close to half of all firms (50% and 49% respectively) expect negative effects given their high dependency on exports to the US (15% and 10%, respectively), which is their top export market. The US (46%) and Vietnam (48%) also record a higher share of firms that expect a negative impact from the trade war compared to the overall sample. Furthermore, when comparing to firms' expectations for 2025, German companies' expectations for 2026 are slightly more negative than they were after "Liberation Day" (+2pps to 49%), while Poland (-5pps to 38%) and UK (-7pps to 41%) expect a less negative outlook compared to pre "Liberation Day". In contrast, firms in Brazil, India and Poland seem to be slightly less worried, with around one-third expecting further negative impacts.

One year on, fewer firms (32%, -6pps) expect to raise their prices due to higher tariffs or trade barriers, back to similar levels seen before "Liberation Day". The firms most likely to raise their selling prices are in the UK (42%, + 4pps from post "Liberation Day"), the US (39%, -6pps) and Spain (35%), whereas only 20% of Chinese firms are considering doing so, -25pps from post "Liberation Day" and -9pps from pre "Liberation Day". The share of firms planning to cut their prices in response to the trade war remained unchanged, with only 14% of firms favoring this option. Vietnam (9%) and Poland (10%) recorded the lowest shares, while Italy and Spain appear to have a relatively stronger foundation, with 19% and 18% appearing to be able to withstand a price cut, respectively. Among companies that declared that they would not change their prices, absorbing the cost remains the least favored option alongside exporting to other markets (24% each) while sourcing from other markets remained the preferred option (25%), though it has lost some of its appeal compared to post "Liberation Day" (-6pps).

Companies appear more ambitious in their investment strategies, with fewer focusing on cost-cutting and more prioritizing capital expenditures in strategic areas. Compared to 2025, fewer firms report holding off investments (15% vs. 22%). Cross-country differences remain relatively modest. Firms in China and Brazil are somewhat more likely to delay investments, whereas firms in Vietnam are the least likely to do so (10%). Overall, responses are fairly balanced across the three main strategies: 26% of firms focus on cost-cutting and operational efficiency, 28% on increasing capital expenditures and 31% on expanding into new business lines. The overall mix is more growth-oriented than in 2025, when cost-cutting was more prevalent (31%, +6pps) and capital expenditure less so (20%, -8pps). At the country level, firms in Vietnam stand out as the most expansionary, with 37% planning to increase capital expenditures and only 10% holding back. This aligns with Vietnam's growing role as an industrial hub within "China+1" supply-chain diversification strategies. Finally, expansion into new business lines is gaining traction among Italian firms (38%, +8pps), and remains the dominant strategy for companies in China (37%, -2pps).

Figure 7: Share of respondents expecting a negative impact from the trade war over the next year



Sources: Allianz Trade Global Survey 2026

Digital transformation and AI adoption clearly emerge as the leading investment priority, cited by 26% of respondents, well ahead of sustainability and carbon reduction initiatives (20%) and supply-chain resilience (18%). More traditional levers such as talent acquisition and upskilling (16%), mergers and acquisitions (12%) and reshoring (8%) rank lower, highlighting a decisive shift toward technology-driven competitiveness. This confirms that AI deployment is no longer experimental but has become a core strategic imperative, with firms increasingly viewing it as critical to productivity gains, cost optimization and long-term positioning.

Cross-country patterns provide additional nuance. US-based firms are relatively less likely to prioritize AI investment (18%), despite operating at the global frontier of innovation and adoption. Instead, they place significantly greater emphasis on talent acquisition (24%), pointing to a maturing digital cycle: with foundational infrastructure already largely in place, the binding constraint is now access to skilled labor capable of deploying and scaling AI effectively. In contrast, China-based firms most frequently cite AI as their top priority (37%), reflecting both the rapid pace of adoption and a continued focus on building out technological capabilities at scale.

Since the start of the trade war in 2025, seven out of 10 firms have taken operational steps to face the tariff increase. The responses cluster into three tiers of adaptation. Inventory building and diversification to new markets (64%), sourcing from new suppliers (63%) and rerouting through third markets (57%). Incoterms changes (36%), the most technical and least visible adaptation, typically reflecting shifts in who bears tariff liability at the border, is the least chosen reaction. As one could expect as the two countries were the most predominant actors of the trade war, the US (27%) and China (31%), followed by Vietnam (30%), were the most likely to have reacted in some way. On the other end of the spectrum, Poland and Singapore both had 40% of their firms conducting business as usual.

Companies in China (69%) were most likely to reroute their products through third markets, followed by the UAE (63%). This is in line with our assessment that 21% of usually US-bound Chinese exports were rerouted through India and ASEAN in 2025. On the other hand, the cost of rerouting seems to have deterred companies in Italy (48%) and Germany (50%). Overall, China also records the highest market diversification rate of any respondent country (75%), followed by the UK and Poland (74% each) and Vietnam (71%). Germany, by contrast, records the lowest diversification rate (52%), alongside the UAE (56%) and India (58%). Furthermore, Chinese companies (75%) have been the most proactive in looking for new suppliers over the past year, once again alongside the US (71%), the UK and Brazil (70% each). Chinese firms with long supply chains are most likely to go towards rerouting to new markets or diversifying to new markets, rather than sourcing from new suppliers. Finally, UAE firms (45%) were most likely to change their Incoterms throughout the past year, while only 26% of the UK firms declared having done so.

In 2026, and as has been historically the case, CIF [1] (31% of importers and 25% of exporters) and FOB[2] (30% and 36%) dominate maritime shipping, continuing to be the most preferred INCOTERMS[3] in the global trade universe, particularly used for B2B transactions. CIF is preferred by buyers seeking convenience, as the seller assumes greater responsibility for shipping logistics – covering freight costs (and any associated rates volatility) and providing insurance up to the destination port – thereby simplifying international transactions, particularly in today's high-risk chokepoints. In other words, the seller streamlines trade for the buyer, especially suitable for those less experienced with international shipping. Companies in Vietnam in particular have shown a strong preference for CIF (45% of importers and 43% of exporters), indicating that Vietnamese firms favor assuming greater control over a larger portion of the logistics chain. Meanwhile, FOB is favored when buyers wants more control over the logistics, as the risk is transferred earlier (at the loading port or port of origin), allowing the buyer to optimize its logistics and negotiate shipping costs. This option was particularly favored by importers in Germany (38%).

1 CIF : Cost, insurance, and freight

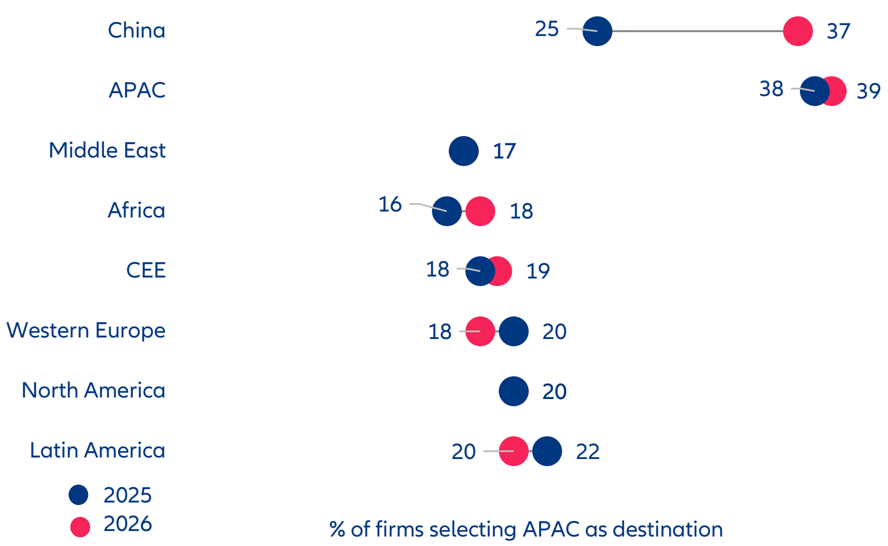
2 FOB : Free on board

3 INCOTERMS stands for International Commercial Terms. They are a set of standardized rules published by the International Chamber of Commerce (ICC) that define the responsibilities of buyers and sellers in international trade

This year's results show clear shifts compared to last year, mainly driven by US tariff policy. Among all 11 INCOTERMS, DDP[4] is unique in placing responsibility for import duties, taxes, and customs clearance on the seller. This helps explain the marked decline in DDP usage by exporting companies, from 25% in 2025 to just 16% in 2026. The shift is understandable, given the trade war environment experienced throughout 2025, which made exporters increasingly reluctant to assume exposure to volatile duties. From the importers' perspective, our global survey shows that the biggest change was observed in more adoption of the FOB term, from 20% in 2025 to 30% in 2026. This shift suggests that importers are increasingly seeking more control over their supply chains, possibly to optimize costs and reduce dependency on suppliers' shipping arrangements.

The Asia-Pacific region excluding China is the clearest structural beneficiary of the current realignment in global supply chains, recording the largest net inflow of any region in the survey. Firms already established in the region show strong conviction in maintaining their presence while firms relocating from elsewhere – most notably from China and North America – identify Asia-Pacific as their primary destination. The pull factors are well understood: competitive labor costs, improving logistics infrastructure, deep regional trade integration under RCEP and maturing supplier ecosystems in Vietnam, India, Indonesia and Malaysia. Singapore's dominant within-region score reinforces its role as the logistical and financial anchor of these value chains.

Figure 8: Firms considering growing in Asia Pacific (excluding China) continue to grow year over year



Sources: Allianz Trade Global Survey 2026

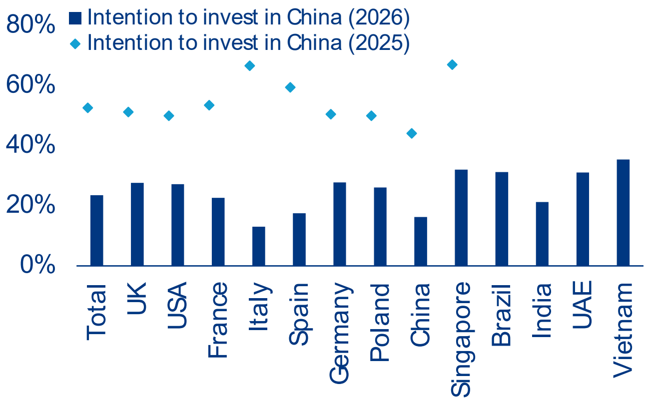
4 DDP : Delivered duty paid

Europe has experienced the largest improvement from last year's survey, while retaining the most within-region stickiness. Interest in Europe as an export destination has grown across the board with Singaporean (+10pp to 19% of firms) and US (+9pp to 29%) exporters showing the strongest increase in appetite, while companies seeking new production sites have also raised their interest in both China and Europe following the Middle East escalation and the start of the trade war in 2025. Notably, the European region records the highest within-region stickiness of any geography in the survey, with approximately 37% of firms currently operating there indicating a preference to remain. Germany and Poland remain the Western European countries most anchored in the CEE region, while Italy is the country with the largest concentration of growth in Western Europe.

The US leads the supply-chain retreat as tariff barriers reduce its attractiveness for businesses. This is not a 2026 phenomenon: Last year's survey already saw a drop in interest in expanding in the US market. This year, only 13% of firms are considering the US as an export growth platform, from 17% the previous year. In parallel, the attractiveness of the US as a potential future location of offshore production sites has continued to deteriorate as the trade war consequences remain. The exceptions are US firms which in this year's survey have turned inward and indicated a higher preference to grow in their own market.

Together, China and the US have suffered the most from the loss of potential future investments, showing how global businesses, as well as Chinese ones, are retreating from the Asian giant. Self-retention among China-based firms collapsed from 32% to 16% while the share of those firms targeting Asia-Pacific as their next location jumped from 25% to 37%, making China towards Asia Pacific the dominant bilateral flow in the survey. Simultaneously, China is losing traction as a destination across almost every non-US region: Middle Eastern firms' interest in China fell from 12% to 4% (-8pps), CEE from 11% to 3% (-8pps), Western Europe from 11% to 5% (-6pps) and Latin America from 8% to 4% (-4pps). In parallel, investment intentions towards China fell from 53% to 24% in a single year — a 29-point collapse that represents the sharpest directional shift in the entire survey. What makes this finding structurally significant is that it is not a Western-led decoupling story: even Chinese firms themselves reduced their pro-China investment intentions by 28 percentage points over the same period, signaling that the reassessment of China as a production base is coming from the inside out, not just driven by geopolitical pressure from abroad.

Figure 9: Investment intentions towards China experience a strong drop



Sources: Allianz Trade Global Survey 2026

While the shift away from China is structural, the Middle East crisis could have introduced some stabilization effects for certain markets that were previously considering to offshore elsewhere. German and UK firms shifted towards a pro-China stance since the beginning of the war, and anchored Singapore's stance even more, most likely as a result of a geopolitical substitution.

Regional cultural gravity continues to shape relocation intent at the bilateral level. Beneath the aggregate trends, the survey reveals persistent home-region bias that cross-regional flows alone do not capture. Spanish firms disproportionately flag Latin America as a preferred destination, reflecting language, legal tradition and historical investment ties. China-based firms, while pivoting to APAC broadly, retain notable interest in Africa and Latin America – destinations where Chinese state-linked investment has built commercial footholds over the past two decades. These gravitational patterns were already visible in last year's survey and have remained stable, suggesting they are durable features of firm-level decision-making rather than responses to the current macro environment, linked to cultural, linguistic and historically established investment corridors.

Companies in countries with recently established FTAs are seeking expansion to new markets, especially in India, Brazil, France and Spain. The sharpest outlier is India, where 40% of respondents want to use FTAs to access fully new markets – the highest of any country by a wide margin. Brazil, France and Spain follow India, pointing towards higher interest to expand to new markets for businesses of those countries actively signing FTAs. China's profile reinforces its defensive posture identified in the relocation data: 13% report no FTA impact – the joint highest alongside Poland – and only 14% are targeting new markets.

Non-tariff barriers remain a challenge but there is a clear split between the difficulties faced by developed and emerging markets. While EMs especially Vietnam (41%) India (37%), China (34%) and Brazil (27%) aim to target and expand in DM markets, the actual barriers remains market-access restrictions: technical standards, anti-dumping duties, phytosanitary requirements, local content rules and import licensing. The EU-Vietnam FTA was built almost entirely to dismantle these for Vietnamese exporters. DM-headquartered firms show a different profile depending on the market. The UK (27%) leads on customs procedures, a consequence of Brexit, while France (24%) also skews customs-heavy, reflecting similar friction for its extra-EU exports. Germany distributes more evenly across state aid/competitive distortions (19%) and quotas (32%). This split is not coincidental as it maps directly onto the direction of trade ambition.

The consensus that broke: sustainability in a globally fragmented world

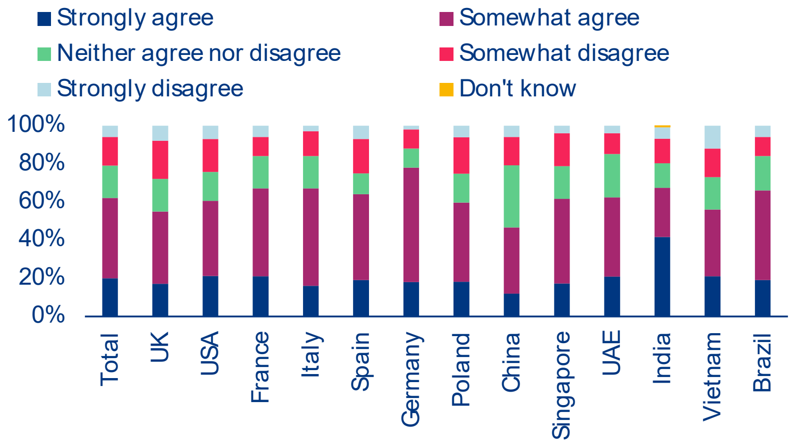
Although sustainability is increasingly shaping global trade, countries are taking divergent paths, creating uncertainty for businesses. For much of the past decade, ESG appeared to be converging towards a unified global standard. This was driven by milestones such as the Paris Agreement, the EU's Sustainable Finance Disclosure Regulation, the launch of the International Sustainability Standards Board and the EU's far-reaching Corporate Sustainability Reporting Directive. A harmonized disclosure architecture was still considered plausible in 2023. However, this changed in 2025 when the US diverged sharply while other regions continued to advance their own frameworks. Sustainability did not disappear but rather fragmented into multiple regulatory regimes with no common baseline. Today, sustainability is increasingly shaping global trade, albeit along divergent national paths that create growing uncertainty for businesses. The incorporation of environmental criteria into trade policy, most notably through border carbon pricing, as exemplified by the EU's Carbon Border Adjustment Mechanism (CBAM), which came into effect in January 2026, is directly linking market access to emissions intensity. This signals a strong regulatory push without any corresponding global alignment. Meanwhile, the US is scaling back certain sustainability-driven trade measures at the national level, while the EU is softening or delaying aspects of its own rules, further widening the gap. Supply chains are at the center of these developments as sustainability requirements now extend across entire value chains, demanding verified ESG data from suppliers. For firms operating globally, this convergence of regulatory fragmentation and trade-linked sustainability increases complexity, raises compliance costs, encourages selective decoupling and heightens trade frictions. The result is an increasingly uneven competitive landscape with risks of regulatory arbitrage and inconsistent market access, making the long-term trajectory of green trade uncertain.

Amid geopolitical risks, firms are scaling back their sustainability ambitions in 2026, focusing on supply-chain measures while deeper internal reforms lag. While sustainability initiatives are being postponed rather than abandoned, firms are prioritising addressing liquidity, geopolitical risk and weaker demand. If cost pressures ease or financing conditions improve, the metrics of the agreement are likely to recover rather than continue to fall. Overall, the results suggest that, compared to 2025, sustainability-related actions will be moderated in 2026, driven less by disengagement than by waning conviction. Although the proportion of respondents who agree that sustainability actions are being pursued remains high, the distribution has shifted, resulting in a net reduction of -22pps in total agreement, from 84% in 2025 to 62% in 2026. This pattern is evident in most countries, though there is significant variation: China has dropped particularly sharply, from 89% in 2025 to 47%, followed by the UK, which has dropped by -29pps, from 84% to 55%. European economies show the least pullback: Germany by -9pps, France by -10pps and Italy by -13pps. Regarding priorities for sustainability measures in business, efforts are primarily driven by supply chains, while deeper internal changes lag behind. On average, supply-chain measures are considered the most important, scoring 59%, particularly in Spain, India and Vietnam (62%), the US and France (61%), compared to 54% in Brazil and Germany, and 53% in the UAE. Logistics and green product investments (39%) are prioritised most highly in Germany for logistics (44%) and in the UK for green products (44%). Governance and compliance actions, such as reporting (33%), regulation (32%) and carbon reduction (31%), remain secondary priorities. The lowest emphasis is placed on structural change, with executive compensation (29%) and workforce diversity (27%) performing poorly across the board.

Global intentions to cut CO_2 emissions are on the rise in 2026, driven by new survey participants and strong gains in India, Vietnam, Poland, Germany and China, while some traditional markets like France, Italy, and the US show slight pullbacks. The intention to reduce CO_2 emissions increased in 2026, with 26% of respondents aiming for a reduction of 5-10% (+4pp compared to 2025). This was driven by ambitious targets in new survey countries such as India (54%) and Vietnam (34%). There were gains in Poland (+8%), Germany (+5%) and France (+5%), while Italy (-7%) and the US (-2%) prioritised sustainability less. Companies' confidence in reaching net zero remains high at 84%, with notable increases in Poland, China, Singapore, the UK and Germany, but declines in France, Spain, the US and Italy. Newly surveyed countries also show strong agreement: Brazil (90%), Vietnam (84%), India (82%) and the UAE (82%).

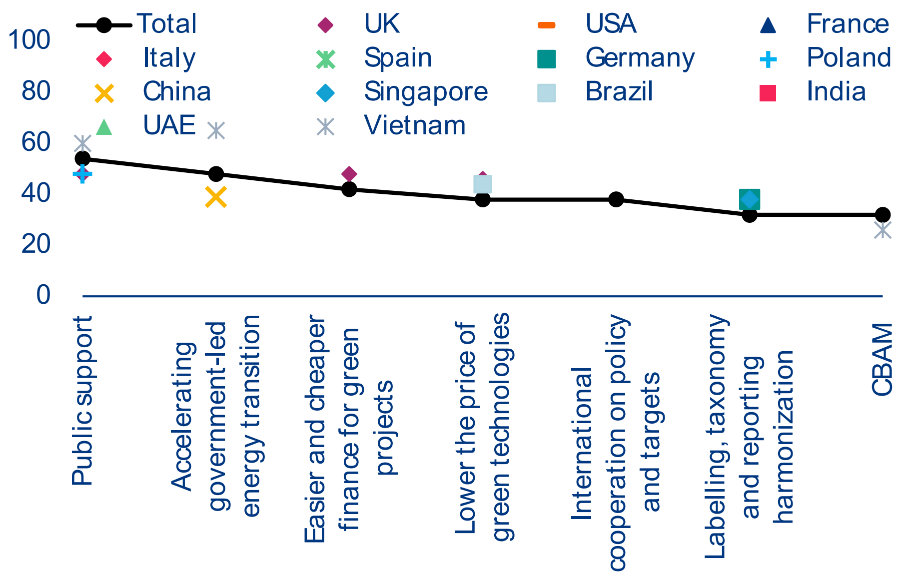
Public support could help firms reduce their carbon footprints. Survey results show that businesses consider public support, such as price subsidies, removal of duties for green products, tax breaks and procurement rules, would have the greatest positive impact on reducing their carbon footprint. This is the most important measure in the US, France, Italy, Spain, Germany, China, Singapore, the UAE and India. This is followed by accelerating government-led energy transitions (e.g. improving the electric grid) with 48% support, leading in the UK (51%), Poland (51%), Vietnam (65%) and Brazil (50%). In contrast, the EU's CBAM is considered to have the least impact, selected by only 32% of companies.

Figure 10 :My company and our senior management implement ESG actions that have significant consequence on our business (i.e. new products, restrictions on clients and suppliers etc.)'



Sources: Allianz Global Trade Survey 2026

Figure 11: Priorities of ESG measures for your business currently, deviation from the mean in pps



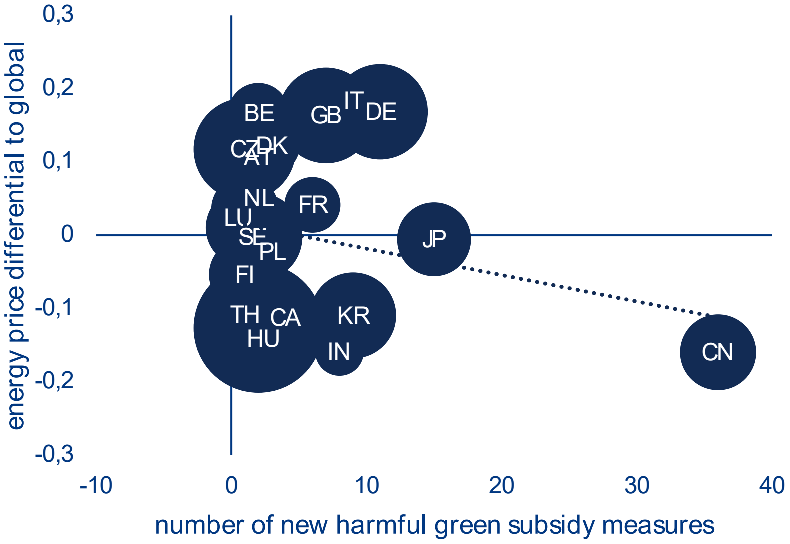
Sources: Allianz Global Trade Survey 2026

Notes: only countries deviating by more than 5pps comapred to the sample average are shown. Public support corresponds to price subsidies, removal of duties for green products, tax breaks and procurement. CBAM stand for Carbon Border Adjustment Mechanism.

Sustainability is becoming central to industrial policy and competitiveness as governments align trade and climate tools, such as subsidies, tariffs and procurement, to support domestic clean industries. However, uneven ambition and shifting regulations, particularly in the EU and US, risk distorting competition. This is especially critical for Germany, where energy-intensive sectors depend on clear policy support during the transition. While about 71% of green goods trade was covered by subsidies in 2025, up from 66.3% in 2024, the number of new measures has dropped from the peak of 367 in 2023 to 235 in 2024 and 190 in 2025. Countries with more new subsidies for green products show a lower energy price differential to global energy prices, while at the same time gaining green export shares. But European countries are holding up strongly in this, despite generally higher energy prices and lower industrial policy support.

Supply chains in the crossfire. Supply chains are where ESG fragmentation is felt most acutely, particularly by mid-tier exporters and Tier 1 and Tier 2 suppliers caught between customers and regulators operating under incompatible frameworks. Firms with deep exposure to the US and Asian markets face disproportionate pressure. A single exporter with operations across the EU, US and Japan must now navigate three non-interoperable systems simultaneously: the EU's CSRD, California's SB 253 and Japan's ISSB-aligned regime. Their underlying logics differ, with double versus single materiality, inconsistent Scope 3 treatment and divergent disclosure formats, forcing firms to maintain parallel data architectures and reporting processes. The cost burden is substantial: Official estimates place annual CSRD compliance for large firms at around EUR740,000, with significant upfront investment, while real-world figures vary widely and can reach into the tens of millions for complex multinationals. Crucially, US competitors often face no equivalent federal requirements, creating a structural cost asymmetry in globally competitive B2B sectors such as machinery, chemicals and automotive supply chains. This gap translates directly into margin pressure, regardless of the underlying quality or strategic value of sustainability data, and embeds a persistent disadvantage for firms operating across multiple regulatory jurisdictions.

Figure 12: Number of new green industrial subsidies relative to energy cost differential to global in costs per kWh



Sources: LSEG Workspace, Allianz Research

At the same time, the EU's attempt to push ESG requirements upstream through supply chains has been weakened, disrupting what was intended as a transmission belt for sustainability standards. The Omnibus reform introduced a value chain cap that limits the ability of large firms to require detailed ESG data from smaller suppliers, many of which form the backbone of Germany's industrial base. This creates a legal and operational gap: companies remain responsible for reporting Scope 3 emissions, which make up a large share of their incorporated emissions in total, but lack enforceable mechanisms to obtain the necessary data. Firms that had already invested heavily in supplier data systems now face uncertainty over the relevance of those investments, highlighting the cost of regulatory instability. More broadly, sustainability has entered a new political economy in which compliance is no longer just a matter of meeting standards but of managing competitive exposure. While cost asymmetries with less regulated markets, particularly the US, can disadvantage European firms in price-sensitive sectors, investor expectations for sustainability performance remain high, and in some markets ESG capability still confers financial and strategic advantages.

The gap between regulatory expectations and operational reality is widening, but the global ESG baseline is rising not falling. Geopolitical fragmentation, driven by US tariff policy, EU supply-chain resilience strategies and a broader de-globalization trend, is prompting European firms to diversify their supply chains, moving them away from China and towards ASEAN, India, Eastern Europe and parts of Africa. This process is creating a new ESG gap. As firms move into supplier markets with weaker, less audited disclosure regimes, they must rebuild ESG data systems for partners who often cannot meet the standards of existing suppliers. This widens the gap between regulatory expectations and operational reality, particularly for companies that are still subject to CSRD reporting. The irony is acute: efforts to increase supply-chain transparency are colliding with geopolitical pressures that are redirecting sourcing towards jurisdictions with less-developed ESG infrastructure, just as EU enforcement mechanisms have weakened. However, this gap may not be permanent. Around 40 jurisdictions representing approximately 60% of global GDP are adopting or aligning with ISSB-based disclosure frameworks, including Japan, Singapore, Hong Kong, Australia and the UK. Meanwhile, China's 2026 CSDS adopts double materiality, which is more closely aligned with the EU. This suggests that the global ESG baseline is rising, not falling, even as the US backs away from federal-level action. For supply chains, this implies a complex transition involving near-term fragmentation and compliance risk, but also a medium-term opportunity to rebuild more consistent and scalable ESG data architectures across increasingly regulated supplier

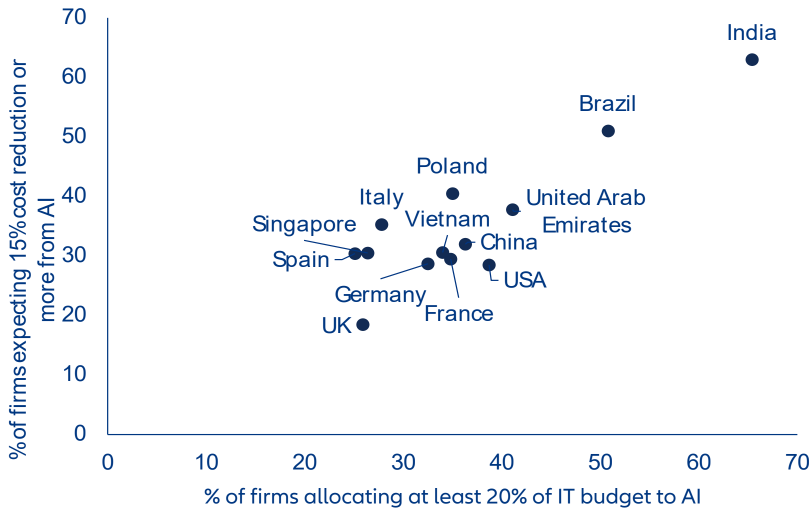
AI adoption among exporters: a two-speed revolution?

AI adoption is now effectively universal among exporters but the depth and strategic weight varies sharply across markets. Only 0.5% of surveyed exporters report not using AI at all, confirming that the technology has crossed the threshold from innovation to daily operations. But while 43% of firms have already deployed AI at scale across multiple functions, only 32% consider it fully embedded in core strategy. The leaders are not where conventional wisdom might expect: China (41%), the UAE (40%) and Brazil (36%) top the rankings for strategic integration. By contrast, advanced economies appear more cautious – only 22% of companies in the UK and 21% in Singapore define AI as central to strategy. Singapore stands out operationally, with 51% of firms achieving scaled deployment, while Vietnam leads globally at 53%, underscoring a growing cohort of emerging-market players that excel in execution even if they remain under the radar.

The real divide is not adoption but belief: emerging markets are far more bullish on AI's near-term impact. The most striking asymmetry lies in expectations, where emerging markets are placing far larger bets. In India, a decisive 63% of exporters expect AI to deliver cost reductions of at least 15% within 12 months – a level of confidence unmatched globally. Brazil follows at 51%, while developed markets lag sharply behind, with just 18.5% in the UK, and Germany (29%), France (30%) and the US (29%) forming a cautious middle ground. The same pattern holds for growth expectations. A sizable 61% of Indian firms anticipate AI will boost export turnover by 10% or more within a year, compared with 30% in Brazil and 29% in the US. At the other end of the spectrum, Spain's 18% signals deep skepticism. This widening expectations gap suggests that emerging-market exporters view AI not as incremental efficiency, but as a leapfrog technology capable of compressing development cycles and reshaping competitive hierarchies.

Spending patterns reflect conviction, but structural barriers – especially ROI uncertainty and talent shortages – continue to constrain expansion. Unsurprisingly, capital allocation follows optimism. In India, 65% of exporters dedicate more than 20% of their IT budgets to AI, a level of commitment that dwarfs most developed markets. Brazil (51%) and the UAE (41%) also demonstrate strong financial backing, reinforcing their strategic intent. Europe, by contrast, remains more restrained: Spain (25%), the UK (26%), and Italy (28%) sit at the lower end of investment intensity. Yet even where spending is robust, scaling AI remains uneven. In advanced economies, the dominant constraint is not access but proof: ROI uncertainty is cited by 35% of firms in the UAE, 31% in Singapore and Germany and 30% in the US, indicating that deployment has outpaced measurable returns. Elsewhere, the bottleneck is human capital. Skills shortages affect 31% of firms in China and India, 30% in Vietnam and 28% in France, highlighting a global scramble for AI talent. Poland stands apart, with 33% citing high implementation costs – a reminder that capital constraints still matter in certain markets.

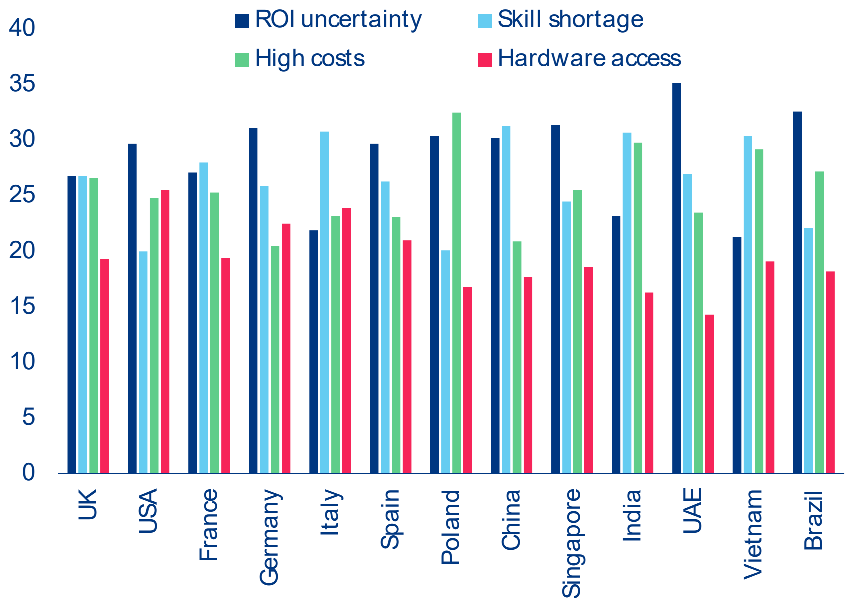
Figure 13: Expectations from AI vs budget allocations



Sources: Allianz Trade Global Survey 2026

A two-speed AI economy might be emerging, with aggressive adopters potentially accelerating ahead – if their expectations materialize. Taken together, the data points to a bifurcated global landscape. Exporters in India, Brazil and the UAE are not only adopting AI – they are committing capital, embedding it strategically and expecting rapid, tangible returns. Their counterparts in Europe and North America, while equally engaged in adoption, are advancing with greater caution, constrained by stricter return thresholds and more incremental expectations. This divergence may reflect differing competitive pressures, baseline productivity levels or institutional risk tolerance. For investors and policymakers alike, the implications are significant: markets exhibiting both high investment intensity and high expectations could deliver outsized productivity gains – or face sharp recalibration if those expectations prove overly optimistic. The next 12 months will be decisive, serving as a real-world stress test of whether emerging-market confidence in AI is prescient – or premature.

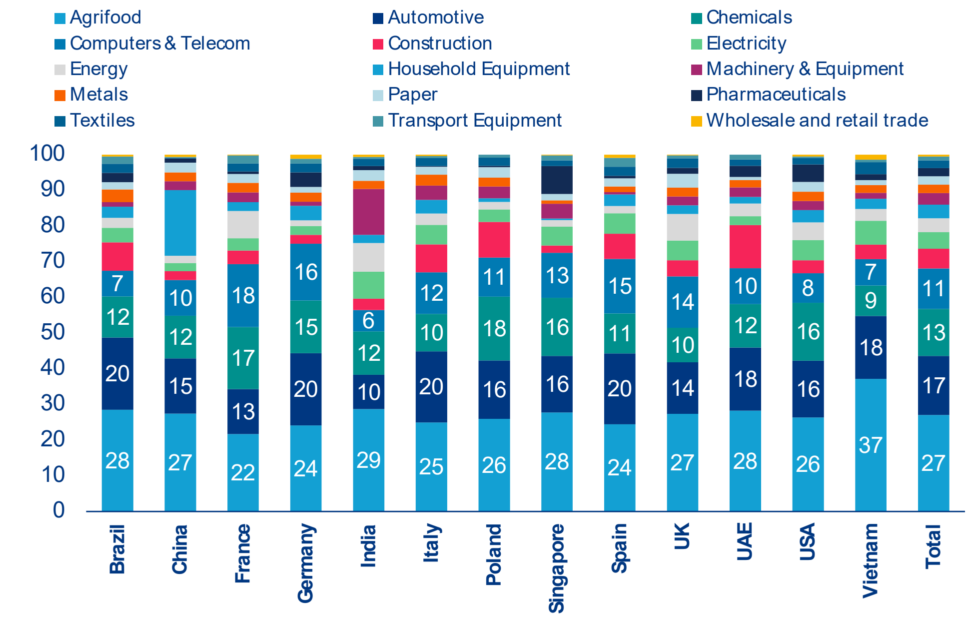
Figure 14: Main hurdles to AI adoption



Sources: Allianz Trade Global Survey 2026

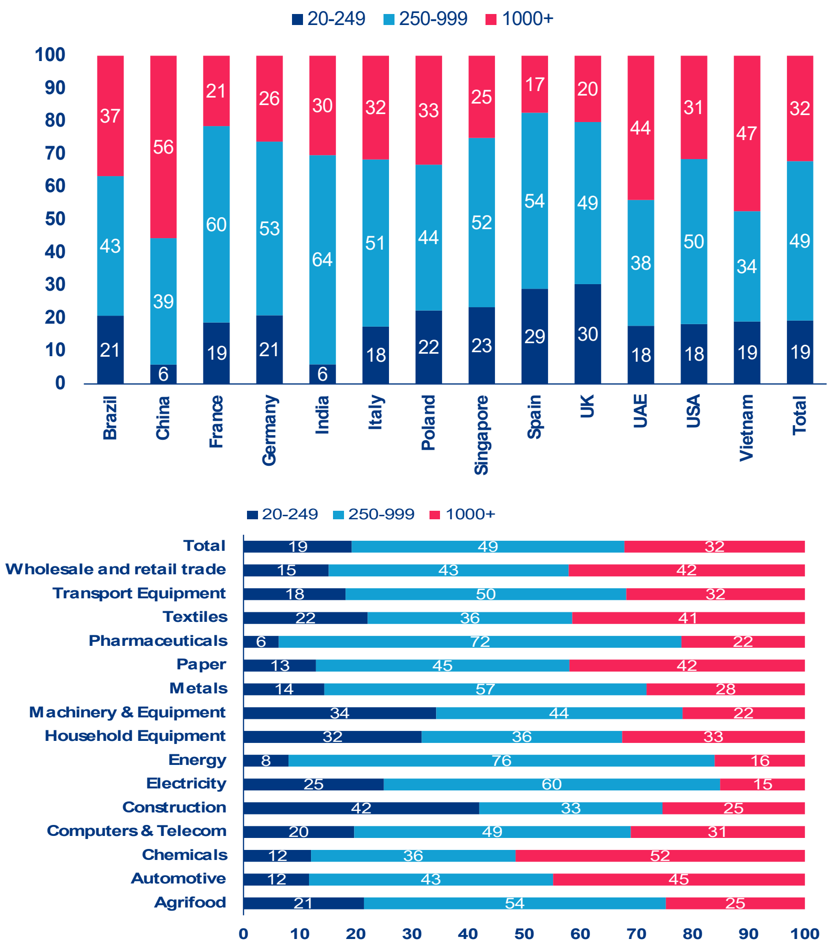
Appendix

Industry distribution across respondents, % of surveyed companies



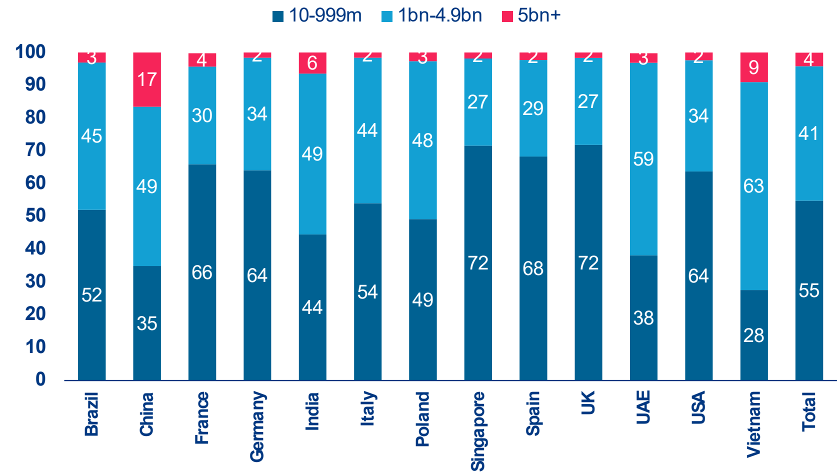
Sources: Allianz Trade Global Survey 2026

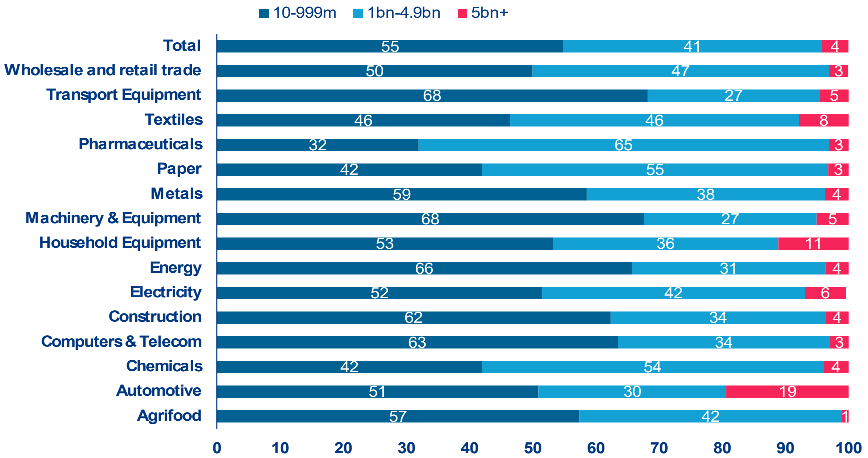
Organization's size distribution by country and sector, in number of employees, % of respondents



Sources: Allianz Trade Global Survey 2026

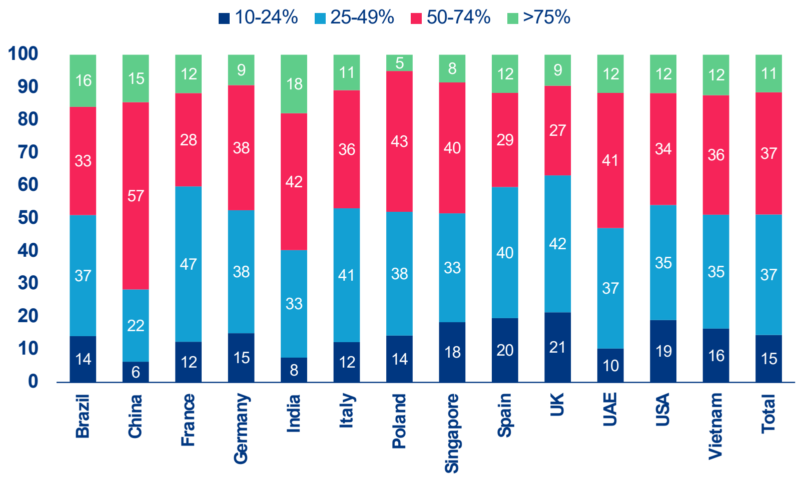
Organization's turnover distribution by country and sector, % of respondents

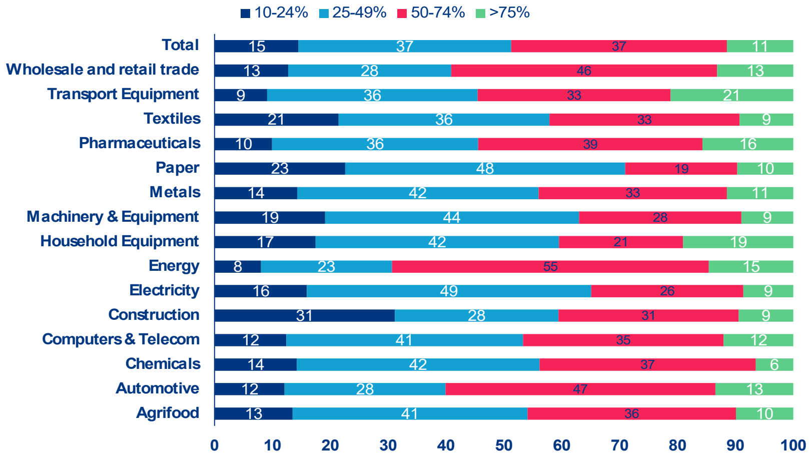




Sources: Allianz Trade Global Survey 2026

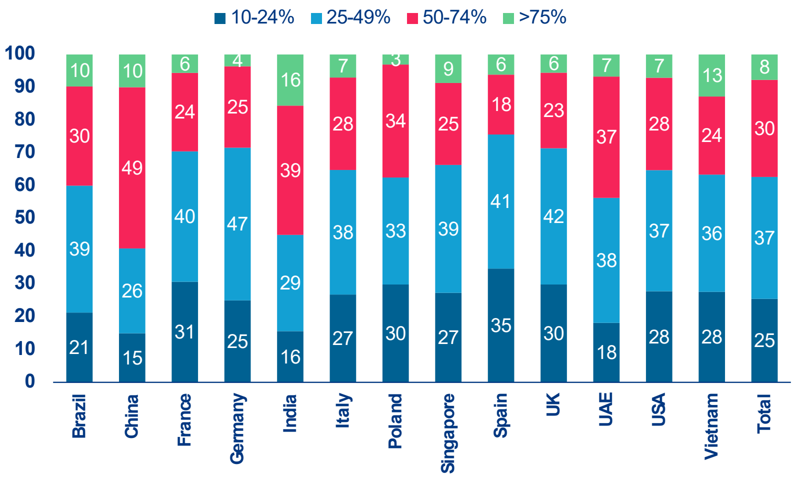
Distribution of companies based on the percentage of turnover generated outside of their company's 'main location', % of respondents

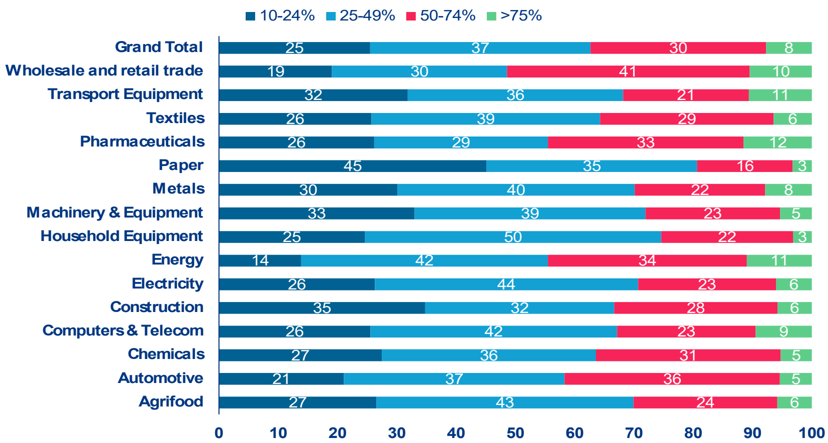




Sources: Allianz Trade Global Survey 2026

Distribution of companies based on the percentage of production (including components) done outside of their company's 'main location', % of respondents





Sources: Allianz Trade Global Survey 2026

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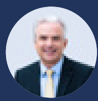

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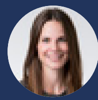

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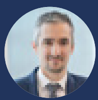

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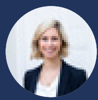

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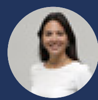

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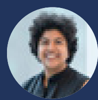

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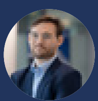

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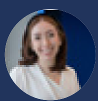

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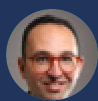

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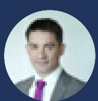

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About Allianz Research
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AI capex cycle: war-proof for now

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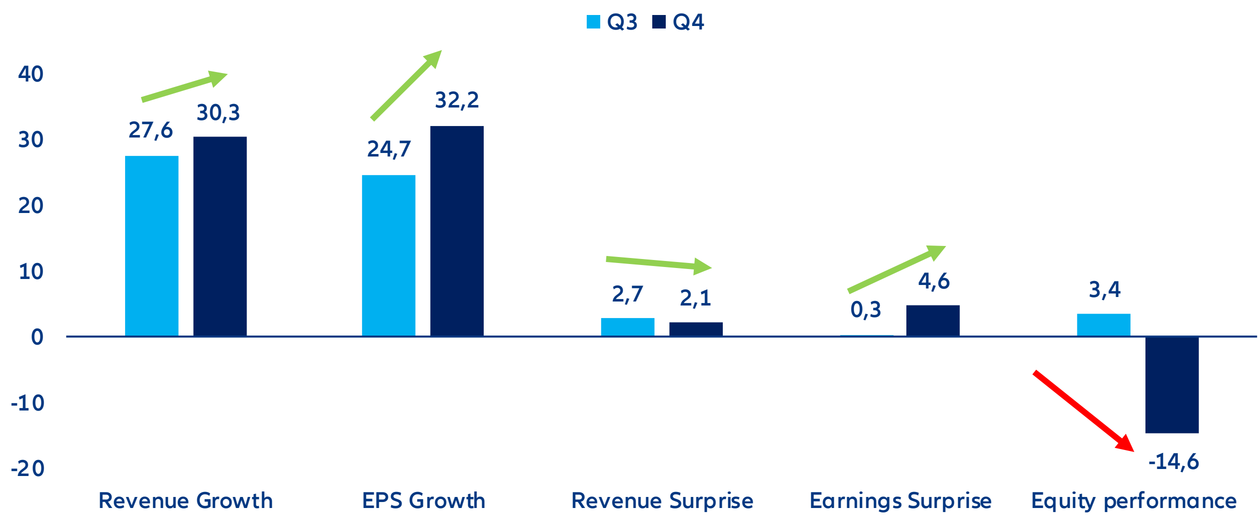
In Summary

- **AI hype deflates amid the capex-monetization debate.** Despite strong recent earnings, investor focus has shifted from profitability boost to revenue growth trajectory and cash-flow visibility, particularly given hyperscalers' elevated capex plans (≈USD 575bn, +50% expected in 2026) and weakening sentiment toward software amid risks of AI-driven revenue dilution. Our AI Bubble Risk Monitor continues to signal moderate bubble pressures: exuberant positioning has cooled, but widening credit spreads send signals of a higher market sensitivity over balance-sheet quality. Rising geopolitical tensions in the Middle East have further reinforced a rotation away from high-valuation tech as risk-off dynamics regain prominence.
- **The higher volatility regime will test the AI development regime, but the capex supercycle remains intact for now, supported by strong, counter-cyclical demand.** Technology capex has historically been sensitive to the macroeconomic environment and notably energy shocks as new inflationary pressure often results in higher interest rates, and consequently higher investment costs. In the US, tech is among the capital-intensive sectors most negatively impacted by severe energy price variations (~-30% correlation) but unlike energy, basic materials and utilities, the drop in investment does not result from windfall benefits stirred up by price effects. Nevertheless, hyperscalers' dominant market positions and substantial cash reserves reduce sensitivity to macroeconomic and geopolitical shocks. In parallel, public-sector investment is accelerating, driven by digital sovereignty and infrastructure build-out agendas. The data-center pipeline is robust (x2 at ~200GW by 2030) and momentum remains resilient despite geopolitical tensions, as illustrated by Germany's plan to double its capacity over the same horizon.
- **Energy volatility may reshape capex allocation rather than overall scale**. While near-term spending levels appear secure, the current concentration in data centers and cloud infrastructure could evolve. AI-related orders benefit from priority access within semiconductor supply chains, limiting immediate exposure to potential disruptions in South Korea and Taiwan linked to LNG and helium supply risks. However, a further ~50% increase in chip costs – as seen in Q1 2026 – could delay project timelines and accelerate the shift toward leasing models to share capital intensity (with estimated savings of 20–30%). At the same time, the case for expanding critical equipment and raw material production outside Asia is strengthening as current tensions expose the risks of supply-chain concentration and may rebalance investment away from the current bias toward software and computing services.
- **How are markets trading the "AI theme"? Focus is shifting from efficiency gains to revenue delivery.** While consensus broadly reflects current momentum, it is increasingly exposed to execution risk on both capex discipline and revenue realization. Valuations – mid-20s P/E for large caps – remain broadly justified but are vulnerable to short-term rotations in a high-volatility environment. The long-term outlook remains supportive, with capex anchored in tangible infrastructure orders that are partly decoupled from the pace of AI adoption. However, value creation is likely to be uneven across the technology stack, with relative winners in semiconductors and telecom equipment, and more challenged segments in software and consumer electronics.

AI cycle: capex monetization debate steals the show during last earnings season

The Q4 2025 earnings season confirmed the fundamental resilience and growth of the global technology sector. The NYSE FANG+ Index – our preferred gauge for tech and tech-enabled growth companies – showed robust financial performance in the most recent quarter. Aggregate results extended on previous strong delivery on expectations, with revenues rising by roughly +17% year-on-year and earnings by around +25%. Each of the leading tech firms exceeded estimates, on average by about 0.5% on sales and 1.6% on earnings across all constituents. Corporate guidance remained constructive, best highlighted by Meta's 5-10% upward revision to its sales outlook. Software companies also highlighted the fundamental potential of AI amid amplified concerns about the potential cannibalization of traditional service lines in areas such as legal, consulting and financial services.

Figure 1: Large-cap tech earnings season compared to past quarter (to end February i.e. pre-Iran war)



Note: Equity performance refers to the three-month period to the end of reporting season and relative to S&P Equal Weighted. Sources: Bloomberg, Allianz Research

Table 1: Earnings & revenue surprise and annual growth among listed US technology firms

	EPS			Revenue		
	Surprise	YoY%	Next quarter rev.	Surprise	YoY%	Next quarter rev.
Computer Hardware	7%	11%	0.0%	3%	7%	0.0%
Computer svs	8%	15%	-2.1%	1%	11%	0.0%
Consumer digital svs	11%	23%	-1.0%	1%	10%	0.0%
Consumer non-digital svs	1%	-11%	-9.4%	0%	15%	-0.4%
Consumer electronics	16%	33%	2.2%	4%	14%	1.7%
Electrical & Electronic Components	7%	32%	0.0%	3%	17%	0.0%
Electronic equip.	4%	8%	-0.1%	2%	8%	0.0%
Tech equip. manufacturing	6%	2%	1.9%	2%	8%	1.9%
B2B Services	5%	14%	-0.6%	1%	6%	0.0%
Semiconductor	6%	29%	0.1%	1%	16%	0.0%
Software	8%	20%	0.0%	2%	14%	0.1%
Telecommunications equip.	12%	19%	0.6%	2%	19%	0.4%
Telecom svs	1%	68%	0.0%	1%	2%	-0.1%
Transaction Processing Services	11%	22%	0.2%	2%	11%	0.2%

Data as of 10 March 2025 based on a sample of 383 companies. Sources: LSEG Datastream, Allianz Research

AI capex is set to continue increasing further in 2026, alongside investors' concerns. AI capex remains the defining market debate in 2026 as technology leaders race to build computing capacity they believe will secure – and deepen – their dominant position in a world where data and processing power are becoming strategic geopolitical assets. After rising ~60% in 2025, US Big Tech investment is expected to climb another +50%, surpassing USD600bn and lifting capex intensity to roughly 23% of revenue, more than double pre-ChatGPT levels. Tech now operates with higher capital intensity than telecom operators, whose investment ratios have declined as innovation shifts toward specialized satellite startups. Yet concerns are mounting: monetization timelines remain uncertain, power infrastructure expansion is lagging data-center demand (utilities capex +20% in 2025, +15% in 2026) and investors increasingly question funding massive spending cycles without the dividends or buybacks traditionally used to compensate capital-heavy business models.

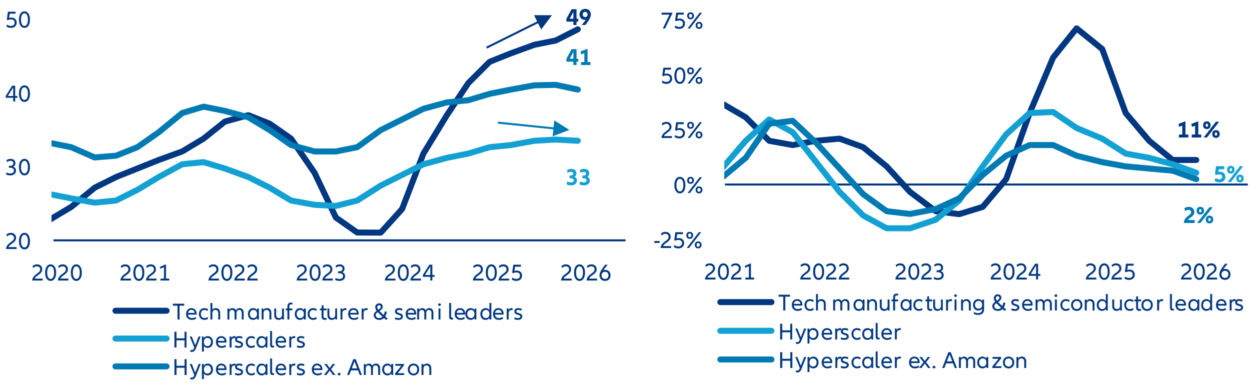
Table 2: Capex growth and intensity over revenue from top 50 biggest spenders within S&P 500 index, per industry breakdown

Industry	End-2023		End-2024		End-2025		End-2026	
	Capex ratio	YoY	Capex ratio	YoY	Capex ratio	YoY	Capex ratio	YoY
Technology*	11%	-4%	14%	44%	19%	58%	23%	50%
Telecom**	16%	-8%	15%	-4%	14%	0%	13%	-5%
Utility***	41%	11%	43%	4%	46%	20%	50%	15%

Sources: LSEG Datastream, Allianz Research

Markets are now questioning how long exceptional margins can last. Investors increasingly doubt the sustainability of profits boosted by quasi-monopolistic positioning and temporary demand–supply imbalances. Early signs from hyperscalers point to easing profit momentum, while Q4 2025 results – despite a +7% median earnings surprise and +17% annual growth – triggered no upward revisions to next-quarter estimates, reinforcing the idea of a profitability cap. Revenue growth remains solid but uninspiring (+2% median surprise vs. +12% for annual growth), reflecting expectations of gradual AI adoption constrained by regulation, costs, trust gaps and skill shortages. As margin expansion normalizes, markets are shifting attention to whether revenue acceleration can compensate, with IT and telecom services seen as most exposed to substitution risks and profit dilution.

Figure 2: Operating margin & profit-revenue growth gap of AI tech leaders* and hyperscalers



**Nvidia (chip design), ASML (chip equipment), TSMC (foundry), SK Hynix (memory chip manufacturer). Data as of Q4 2025. Sources: LSEG Datastream, Allianz Research*

Our AI Bubble Risk Monitor continues to hint at moderate bubble pressures, though the underlying drivers have changed to some extent. Fundamental developments around adoption momentum, data center vacancy rates and GPU product-cycle dynamics remain reassuring. Energy-market developments need watching but see no increase in risk. Meanwhile, exuberant expectations have deflated somewhat reducing the risk of an abrupt market setback triggered, for instance, by an earnings disappointment. In fact, equity positioning, the bear market indicator and 12m forward PE have all come down to more moderate levels over the past three months. Financial risks are on the rise – albeit moderately – and market signals are now flashing red when it comes to US investment grade tech spreads, which have recently moved to bubble-like territory, likely reflecting AI challenging software firms. But overall, bubble risks look less pronounced compared to a month ago. At this stage, tail risks over the long run remain contained but some shadow areas on the pace of capex monetization and rapid cash balance deterioration require monitoring, especially because capex is expected to continue increasing at a rapid pace this year, and because imbalances generated by this massive flow of funds could be broader than cashflow.

Table 3: AI Bubble Risk Monitor

	Indicator	Q1 2024	Q2 2024	Q3 2024	Q4 2024	Q1 2025	Q2 2025	Q3 2025	Q4 2025	Latest (Feb 25)
Market Signals	12M price return	2	1	1	1	0	0	0	0	1
	S&P 500 12M yoy breadth	0	0	0	0	0	0	0	0	0
	Data Center Spreads	0	0	0	0	0	0	0	0	0
	US IG Tech spread	0	0	0	0	0	0	0	0	0
	US HY Tech spread	0	0	0	0	0	0	0	0	0
	US HY Tech vs HY spread	0	0	0	0	0	0	0	0	1
	US IG Tech vs IG spread	0	0	0	0	0	0	1	2	3
	S&P 500 1Y implied vola	0	0	0	0	0	0	0	0	0
	10Y Treasury Yield	1	2	0	1	2	1	1	0	1
	S&P 500 Put/Call-ratio	0	0	0	0	1	0	0	0	0
Exuberant Expectations	Long-term earnings growth	0	0	0	0	0	0	0	0	0
	Earnings revisions (3M)	3	3	3	3	3	3	3	3	3
	Tech EPS forecast dispersion	0	0	0	0	0	0	0	1	1
	EQ positioning	1	0	0	1	0	0	1	2	0
	Bear market indicator	0	0	1	2	3	3	3	3	2
	12M forward PE	1	1	1	1	1	0	1	1	0
Financial Risks	Net-debt-to-equity ratio	0	0	0	0	0	0	0	0	0
	Cash on balance sheet as % of market value	0	1	1	1	1	1	1	1	1
	TED spread	0	0	0	0	0	0	0	0	0
	Cryptocurrency volatility	1	0	1	1	0	0	0	0	1
	Hyperscalers CapEx/Cash flow from operating activities	0	0	0	0	0	1	1	2	2
	US 5Y5Y Forward Expected Inflation	0	0	0	0	0	0	0	0	0
Fundamentals	Current Usage of AI	1	1	0	1	0	0	0	0	0
	Expected Usage of AI	2	1	0	0	0	0	0	0	0
	DC power demand vs generation supply growth							2	2	2
	Power capacity queues	1	1	1	1	1	1	1	1	1
	Electricity prices	0	0	0	0	0	0	0	0	0
	Operational Health	0	0	0	0	0	0	0	1	0
	Capex-to-sales level	1	1	1	1	2	2	2	2	2
	Data center pipeline (construction vs. commited)	0	0	0	0	0	0	0	0	0
	Data center vacancy	0	0	0	0	0	0	0	0	0
	Product cycle of GPUs	1	1	1	1	1	1	1	1	1

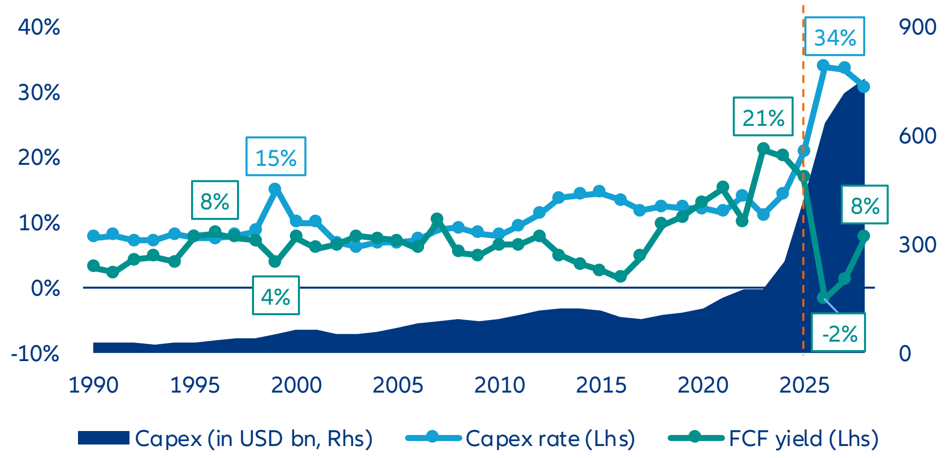
Legend: normal | elevated | overheated | bubble-like

Source: Allianz Research. Note: In addition to general market indicators, where applicable we also monitor AI sub-segments via equity baskets. The monitor assigns scores to each signal 0 = normal, 1 = elevated, 2 = overheated and 3 = bubble-like to help aggregate and compare diverse developments within and across dimensions. Thresholds for scores are derived statistically where long-time data series are available but also via expert judgement

Investment cycle: resilient to geopolitical tensions but watch for an allocation shift

Holding the capex intensity no matter what. The five largest US technology spenders are on course to commit over USD600bn in capital expenditure in 2026 alone – part of an estimated USD2.1trn deployment across 2026–2028 – driving capex intensity to 34% of revenue, more than double the 15% peak seen at the height of the 1990s internet buildout. Figure 3 tells the story starkly: absolute capex is now in vertical ascent, free cash flow is turning negative for the first time in 35 years and the inflection is sharper than anything in the modern history of listed US firms. Growth in capital spending will decelerate – from 51% in 2026 to 13% in 2027 and roughly 5% in 2028 – but that deceleration reflects the natural digestion of an extraordinary base, not a change of conviction. These firms are deliberately engineering a year of negative free cash flow because they have concluded that the revenue potential of AI justifies the sacrifice: technology penetration into enterprise and consumer segments is inherently gradual, and the returns on infrastructure laid today will accrue over years, not quarters.

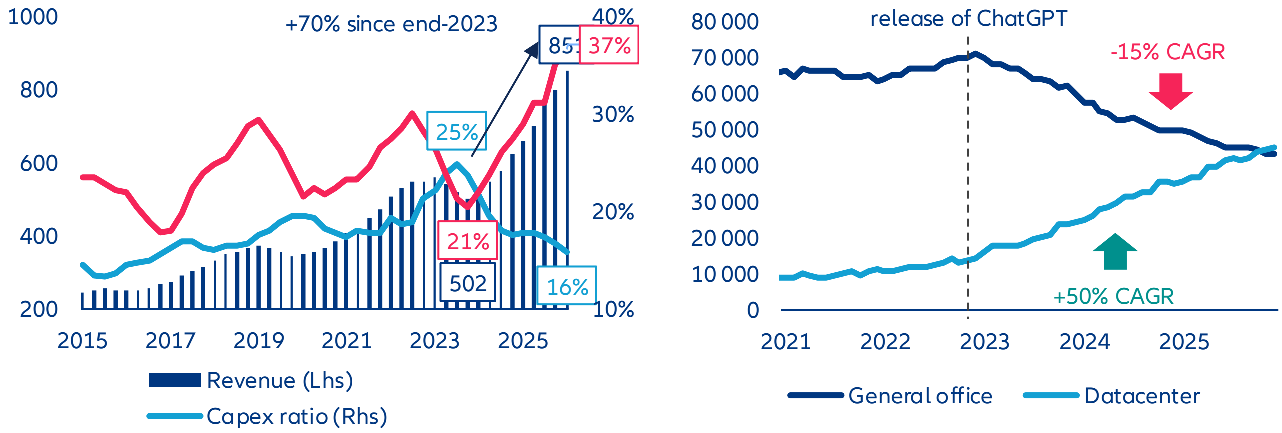
Figure 3: Capital expenditure (amount and intensity over revenue) and free-cash-flow yield of top five US capex contributors among listed firms



Note: The composition of top five capex leaders changes every year. Sources: LSEG Datastream, Allianz Research

AI benefits are already visible, but not yet broad-based. Hyperscalers are committing hundreds of billions of dollars to a technology cycle whose return horizon remains fiercely debated, and skepticism is rational. Yet, reducing this debate to a binary verdict – productive investment or reckless speculation – misses what the data already shows on the ground. US construction spending on data centers has exploded since 2023, while office construction continues its structural decline, a physical reallocation of capital that is reshaping the built economy in real time. Semiconductor revenues tell a similar story: Industry sales surged roughly 70% from end-2023, with operating margins expanding sharply even as capital expenditure intensity hit multi-decade highs. These are not paper gains – they reflect real backlogs, real pricing power and real profit flowing through the supply chain. The distribution of those gains is, admittedly, narrow: A handful of firms with defensible technology edges – in advanced logic, memory bandwidth, cooling and power infrastructure – are capturing a disproportionate share of the AI cake. But the notion that AI investment is generating only financial abstraction, widening inequality without creating tangible economic output, is countered by capacity-utilization rates, order books and the income statements of those positioned at the infrastructure layer. The more pressing question, then, is not whether AI capex is producing returns – it clearly is, for some – but whether those returns are being recycled in ways that build a genuinely durable and expansive ecosystem, or whether they are accruing narrowly and compounding existing structural asymmetries.

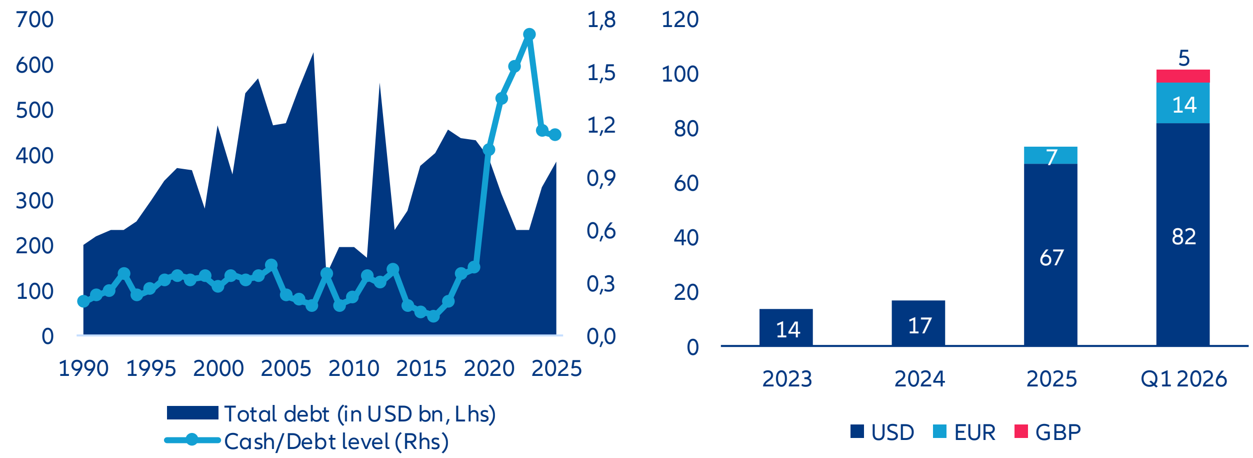
Figures 4 & 5 : Global semiconductor industry revenue, margin and capex intensity / Value of private construction spending in general office and datacenter in USA (annualized rate, in USD bn)



Right chart: November and December 2025 are preliminary data. Sources: US Census bureau, LSEG Datastream, Allianz Research

Solvency risks are overdone but rising debt still needs to be monitored. The parallel with heavy telecom investment in the late 1990s is imperfect as that was largely debt-funded and ended in widespread capital destruction; this round is being funded from the strongest corporate balance sheets in history, by firms with demonstrated monetization engines. But while solvency risks are not a major concern, we cannot overlook the ongoing shift of funding architecture and the increasing volume of AI-labeled debt since 2025. Even among hyperscalers who ignored the debt market for a decade, we note a structural pivot that is gaining more momentum in 2026 as they had already issued as of mid-March more debt than during the full year of 2025 (over USD100bn vs. ~USD80bn). With a total of USD385bn of debt reported in end-2025, the top US capex providers show a leverage ratio around -20% below their "high spender" peers in 2000, so ongoing rising leverage is not a matter of concern in the short term. However, the uptrend will be scrutinized by investors, alongside the downtrend on free-cash-flow.

Figures 6 & 7: Cash ratio & total debt level of top five largest US capex contributors / Corporate debt issuance of US hyperscalers* since 2023

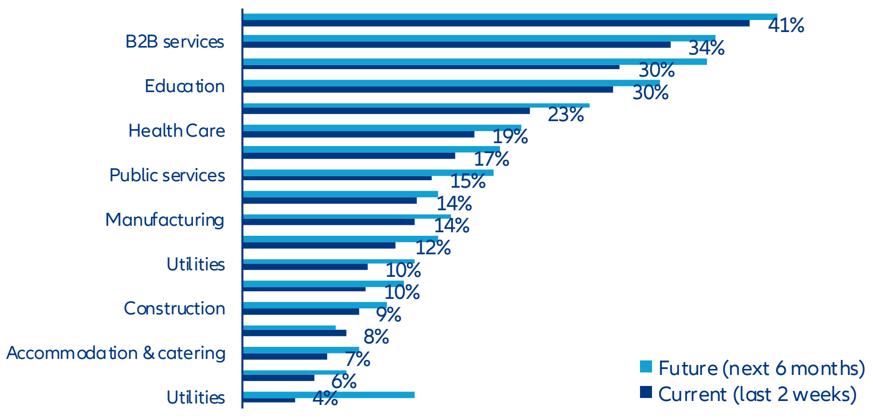


Left chart: data as of Q4 2025 / Right chart: data as of 18 March 2025. Sources: LSEG Datastream, Allianz Research

Beyond the noise: what actually shapes the AI capex cycle. The debate around AI investment has been too often reduced to market valuation stories – a narrative of soaring multiples, crowded trades and the occasional correction fueling breathless commentary about bubbles and busts. The more consequential discussion is industrial, structural and slower-moving. Four fault lines are quietly shaping how – and whether – current levels of AI capital expenditure can be sustained and productively absorbed. First, the pace of corporate and consumer adoption, which is tracking a natural learning curve rather than the vertical inflection that capital deployment implicitly assumes – a structural lag with direct consequences for return horizons. Second, a growing concern around misallocation: the breakneck pace of AI infrastructure buildout is not matched by equivalent investment in the underlying industrial capacity – power generation, grid infrastructure, advanced manufacturing – needed to underpin it, creating tangible shortage risks and inflating input costs across the supply chain. Third, a question of circularity: whether AI profits are genuinely creating new economic value or merely consolidating the dominance of a handful of firms, raising the specter of systemic spillover and contagion risks should any major actor stumble. Finally, new tensions in the Middle East, where energy supply disruptions could reignite inflationary pressure, complicate the rate outlook and dent the investor confidence on which this investment cycle critically depends. Taken together, these fault lines do not make the AI capex story one of inevitable disappointment but they do make it one that demands far more industrial rigor and strategic honesty than the market narrative has so far supplied.

1) **Gradual adoption is fueling monetization debate, not calling into questions capex itself**. The professional adoption of AI remains gradual, exploratory and uneven – and that is not necessarily a bad signal. Corporates broadly recognize AI's long-term potential, yet the transition is proceeding cautiously for structurally sound reasons: firms still need clearer visibility on implementation costs, measurable productivity gains, cybersecurity risks and an evolving regulatory framework before committing to large-scale deployment. Psychological barriers compound the technical ones as companies remain reluctant to build deep dependency on providers lacking long operational track records. The result is that most organizations are currently engaged in incremental process improvements rather than transformative reinvention – a pattern entirely consistent with how major technological shifts have always unfolded. Early gains are predictably concentrated in digitally mature sectors such as consulting, finance and insurance, while construction, utilities or agrifood show only limited impact so far. Financial transfer costs further widen this gap, with adoption among very small enterprises running nearly 10pps behind large corporates. Rather than the binary outcome between automation and employment destruction that dominates media narratives, AI is more likely to act as a complementary technology enhancing existing tasks and business models – with capex implications already shifting visibly toward staff training, compliance, cybersecurity and data infrastructure, the unglamorous but essential prerequisites for durable migration. For technology companies, this pace differential between capital deployed and returns materializing is not merely an investor-relations inconvenience – it is a strategic constraint that demands greater investment discipline and a genuine rethink of how financial risk is shared across the ecosystem. A measured adoption curve is not evidence of deterrence; it is the bedrock on which a solid and sustainable industry gets built.

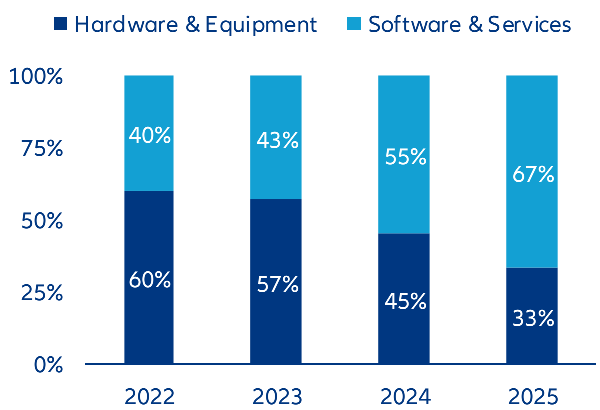
Figure 8: Survey of US firms about their current and expected use of AI technology, per sector breakdown



Data as of 22 February 2026. Sources: US Census bureau (biweekly survey on business trends and outlook), Allianz Research

2) **AI cannibalization of global supply raises question on industrial capacity capex lag**. A disproportionate share of global supply is being absorbed by the AI value chain – particularly for GPUs, high-bandwidth memory, advanced substrates, power electronics, silicon wafers and electronic manufacturing equipment. The result is tightening availability and rising input costs for other sectors. Memory chip prices, for instance, have more than doubled since last summer (see Figure 8) as limited fabrication capacity has been redirected almost exclusively toward AI-oriented demand, creating pass-through inflation for downstream customers and, in some cases, tangible shortage risks. Consumer electronics manufacturers, robotics and IoT producers, and especially the automotive industry – where software and semiconductor content per vehicle has surged in recent years – appear particularly exposed to these side effects. Higher component costs would compress industrial margins and/or raise final retail prices, potentially dampening end-demand for tech hardware. We estimate that manufacturers would have to increase retail prices by +7% for smartphones and +15% for premium laptops to protect half of current profit margins in case of doubling memory chip input costs. In turn, elevated equipment premiums may slow consumer adoption cycles, paradoxically constraining the diffusion of AI-enabled devices. While current AI usage among households remains largely concentrated in writing and search applications, scaling the technology into more embedded, hardware-intensive use cases requires affordable, dedicated equipment; inflationary pressures along the supply chain risk delaying that broader penetration and limiting the real economy spillovers that AI proponents anticipate. If we observe some lag in the capex programs of ICT equipment manufacturers, most of the capital is being captured by data-center development programs as industrial priorities are primarily focused on computing-capacity expansion (current 33%/67% allocation of capex between ICT hardware & manufacturing and ICT services – see Figure 9). Over the next two years, we see a change of capex structure and a stronger allocation of technology-related investment funds into new ICT hardware and equipment industrial capacity, narrowing the capex structure gap with digital & IT services toward a 50/50 ratio, like in 2024.

Figures 9 & 10: Memory chip spot price in USD (Flash NAND 64GB) / Capex distribution between technology hardware & equipment manufacturers and digital & software services providers

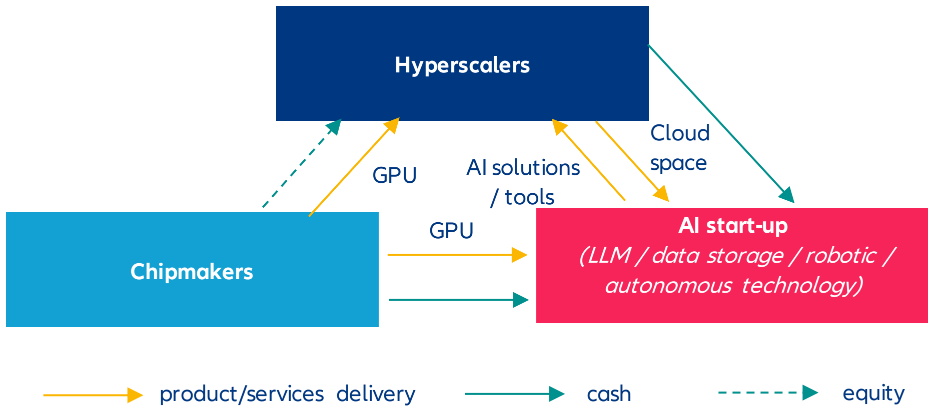


Sources: LSEG Datastream, Allianz Research

3) **Capital circularity casts doubts about the productive role of capex**. In the meantime, circular capital is increasingly dominating the AI ecosystem, especially across chips and cloud, creating a powerful – but potentially fragile – perception of strength along the entire supply chain. Semiconductor leaders such as Nvidia, TSMC or ASML sit at the core of this expansion, riding a +60% CAGR industry revenue surge since end-2022 and benefiting from multi-billion-dollar hyperscaler contracts, 30%+ CAGR cash accumulation and a global market revenue that could hit a USD1trn milestone this year if strong demand remains in check. Yet, beneath the surface, inventory growth among the top US chipmakers (+80% annually) is outpacing already explosive revenue gains, while part of demand stems from structurally unprofitable AI players reliant on external funding, raising questions about the durability of the cycle.
More concerning is that this revenue boom has not translated into proportional industrial reinvestment: capex intensity among the top five global foundries slowed last year to a five-year low at 24% (and hardly 2% for the top three US chipmakers), or almost 10pps below its 2023 peak. In the meantime, capex intensity

among top US designers stayed relatively flat at 2%, with profits being distributed to shareholders, invested in start-ups in early stages of development or directly to business partners to fund new data-center infrastructure. With hyperscalers and tech service providers now absorbing roughly two-thirds of total AI spending, capital allocation appears heavily skewed toward data-center infrastructure, exposing a systemic imbalance. Behind circularity stories, there is a deeper issue within the AI ecosystem: a strong capital interconnection between each key actor of the chain whose funding has been gradually outsourced to financial markets and private investors. That holds seamlessly as long as the AI monetization narrative holds. The moment investors seriously question the timeline – and the friction-heavy reality of enterprise technology adoption gives them every reason to – liquidity assumptions could crack across cash, credit and equity simultaneously, with spillover effects that would reach well beyond the technology sector.

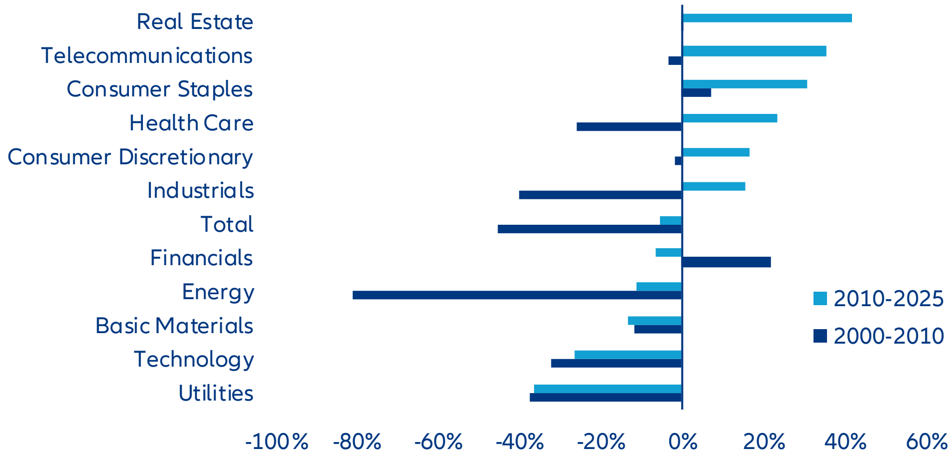
Figure 11: Circularity of investments from the AI ecosystem



Source: Allianz Research

4) **Middle-East tensions revive bad memories: a double chokepoint to monitor.** The re-escalation of Middle East tensions introduces two distinct but mutually reinforcing pressure points on the AI investment cycle – one financial, one industrial – neither of which the market has fully priced in. On the funding side, an energy-driven inflationary shock would complicate the rate outlook at precisely the wrong moment. Under already-intense scrutiny over capex monetization, higher-for-longer rates would give investors and lenders legitimate cover to tighten standards, demand stronger return guarantees and quietly stall the most speculative infrastructure projects. Given the extreme capital intensity of AI – where a single hyperscaler data center can absorb several billion dollars before generating a dollar of revenue – even a modest repricing of risk appetite could trigger meaningful investment pauses across the pipeline. This is not a theoretical sensitivity: the correlation between energy price volatility and technology sector capex in the US is among the strongest across all S&P 500 industries over both the 2000–2010 and 2010–2025 periods, suggesting that energy shocks reliably translate into investment hesitation across the tech supply chain. The silver lining, if there is one, is that capital discipline tends to accelerate efficiency: constrained funding historically pushes R&D toward leaner architectures, and the current cycle would likely see inference optimization and cost-efficient model deployment prioritized over brute-force training at scale.

Figure 12 : Correlation between capex and energy prices among US firms, per industry breakdown (S&P 500 benchmark)

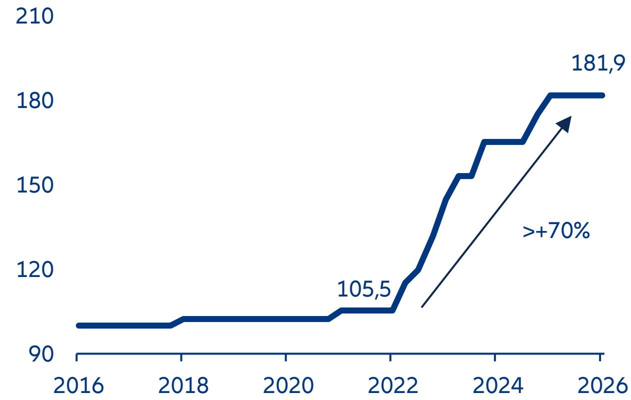


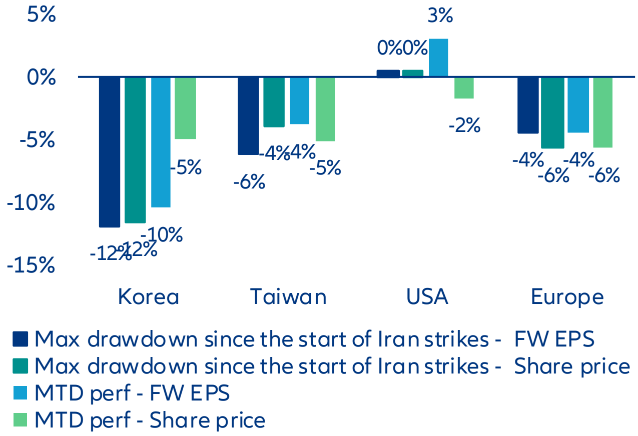
Data as of March 2026. Sources: LSEG Datastream, Allianz Research

The industrial pressure point is more immediate and more concrete. South Korea's industrial power price has surged over +70% since 2022, and both South Korea and Taiwan – home to the world's most critical semiconductor fabrication capacity – remain structurally dependent on Qatari LNG and helium supply as primary inputs for producing wafers. Any supply disruption or price spike flowing from Middle East instability hits Asian chip manufacturers directly in their cost base, compressing margins and risking output constraints at the worst possible moment for an AI ecosystem already running on tight GPU supply.

Is the capex cycle at risk from a chip/critical material shortage? At this stage we consider that risks of disruption in the super investment cycle are limited for two reasons: 1) capex is mostly driven by cash-rich and counter-cyclical companies whose revenue is less dependent on the macroeconomic outlook and 2) AI-related demand remains the top priority across the semiconductor supply chain. Regarding that second point, even moderate disruptions in wafer output are unlikely to materially derail ongoing capex dedicated to AI infrastructure. However, such constraints would limit any near-term acceleration in investment. They could also delay project timelines and increase implementation costs. However, a more volatile market environment will likely reinforce investor caution, particularly regarding datacenter monetization and the visibility of sustainable AI-driven revenue growth. The geopolitical stress does, however, carry one structural positive: it intensifies the urgency of supply-chain diversification, opening a credible investment case for semiconductor capacity in Europe and the US – a rebalancing that could, over time, reduce the systemic concentration risk that currently makes the entire AI buildout hostage to a handful of facilities in a narrow geographic corridor.

Figures 13 & 14 : Industrial power price in South Korea / EPS & share price variation of tech stocks since 27 Feb.

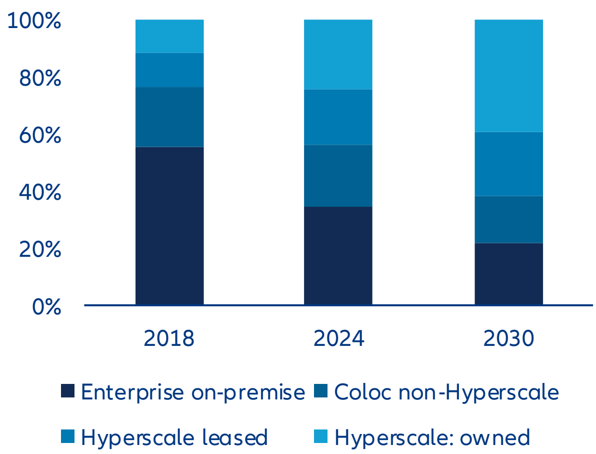




Data as of March 2026. Sources: Public statement from KEPCO, LSEG Datastream, Allianz Research

Smaller but smarter: monetization debate might lead further rationalization. Slowing AI adoption, compressing returns and volatile markets will inevitably test the conviction behind current capex programs. Yet reducing investment is not a credible option: pulling back signals retreat, cedes ground to competitors and risks losing the momentum that separates infrastructure leaders from fast followers. The capex expansion continues – but its risk profile is quietly being renegotiated. The scale of what is being built is extraordinary. Global data center capacity is on course to double by 2030, from roughly 130 GW today toward 260 GW, with hyperscalers as the overwhelmingly dominant engine – their share of total IT load expanding nearly 20pps to exceed 60%. At a 23% capacity CAGR, that implies approximately 100 GW of net new hyperscale deployment over five years. Applying current AI GPU cost structures and next-generation energy intensity assumptions, GPU procurement alone runs at roughly USD16bn per GW – implying a gross annual capex requirement approaching USD960bn if hyperscalers were to fund every megawatt themselves. But they won't bear that financial burden alone. Assuming current leasing ratios hold – where hyperscalers procure IT equipment directly but lease shell and core infrastructure – the annual capital commitment compresses to approximately USD730bn, a figure that aligns squarely with consensus estimates of USD760bn for 2028 alone. The next strategic frontier is pushing that ratio further: higher leasing penetration, more aggressive cost-sharing with co-investment partners and procurement structures that strengthen the AI service arsenal while keeping the balance sheet breathing.

Figures 15 & Table 4 : Distribution of current and future data center IT power load capacity per company profile breakdown / Simulation of yearly capex requirement from hyperscalers over the period 2026-2028 to fund datacenter estimated expansion



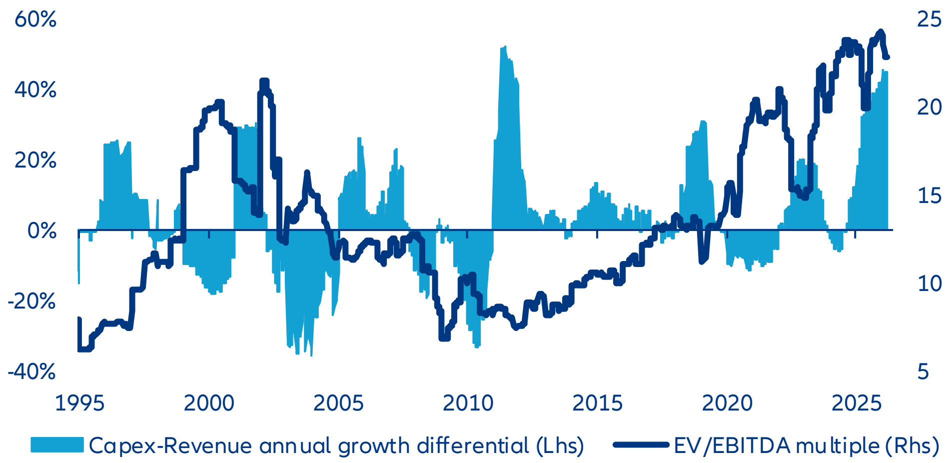
GPU ratio in total DC construction cost / DC full ownership ratio	40%	30%	Average
100%	827	1102	964
75%	703	909	806
63%	645	819	**732**
60%	628	793	711
50%	579	716	647
40%	529	639	584
30%	479	562	521

Sources: Synergy Group Research, Allianz Research

The changing economics of Big Tech: AI investment, profitability and valuation

Too highly valued? Capex monetization is the new compass for market mood. Despite the recent correction, US technology and AI equities remain richly valued relative to earnings, with EV/EBITDA multiples near 25x, close to historical extremes and above telecom valuations preceding the 2000 dot-com peak. The core issue is no longer AI potential but timing: capex is expanding far faster than revenues, with a ~46% growth gap between investment and sales, exceeding the 32% divergence observed during the 2001 telecom excess cycle. Infrastructure is scaling ahead of monetization, forcing investors to question whether current valuations already price in profits that remain distant. As early expectations of rapid AI earnings acceleration fade, markets are recalibrating toward execution risk and cash-flow delivery. This cooling of sentiment may ultimately mark a healthy transition – from narrative-driven valuation to discipline anchored in tangible returns.

Figure 16: Relative valuation and growth differential of revenue and investment from US technology industry

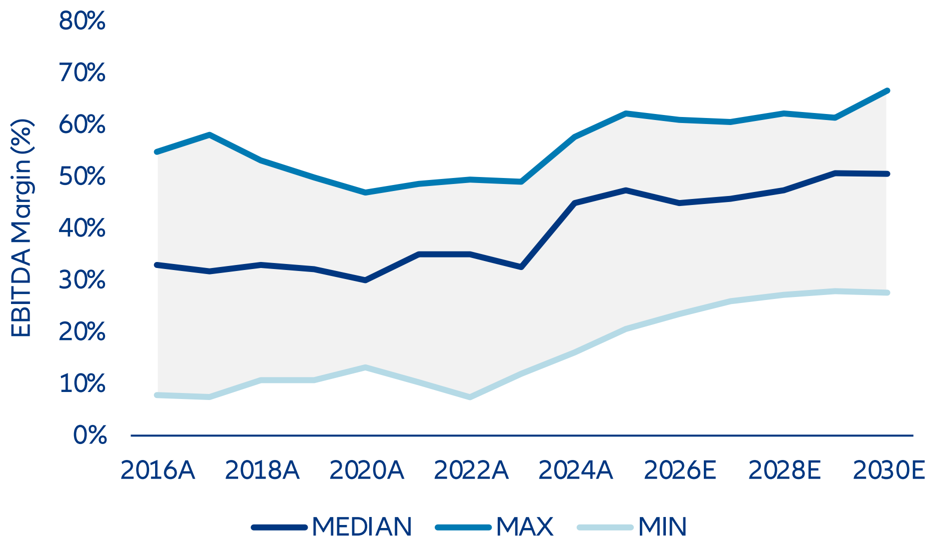


Data as of 16 march 2026. Sources: LSEG Datastream, Allianz Research

What are investors in Big Tech implying for AI growth and profitability? Assessing the profitability of AI-related activities remains challenging at the current stage of development. We are not trying to forecast the growth and profitability of AI companies but to form an opinion on legacy business contribution vs. AI expectations. We are also not trying to assess individual winners or losers but gauge sensitivities of the AI-complex to key assumptions. As a starting point we take consensus and past dynamics of legacy businesses. Segment disclosures and SEC filings (Forms 10-K and 10-Q) can be used to model legacy business activities separately, enabling a consistent projection of underlying operating performance. Given ongoing uncertainty surrounding AI monetization, particular emphasis is eventually placed on sensitivity analysis across key financial variables, with return on equity (ROE) serving as the central link between operating performance and valuation outcomes.

The changing capital intensity increasingly shapes the outlook for profitability. We focus on ROE which best combines the trade-off between capex and monetization. As illustrated in Figure 15, EBITDA margin development suggests that the expansion phase observed in previous years is gradually moderating. Amazon is an outlier at the lower bound, reflecting the still significant contribution of its lower-margin retail business, while hyperscalers remain influenced by the ongoing capex expansion cycle. Profitability remains structurally elevated, yet incremental improvements appear limited. This reflects the structural characteristics of large technology companies, many of which already operate with comparatively low cost bases and optimized operating structures. In fact, somewhat higher EBITDA margins are to be expected for a more capital-intense business and still imply lower net profitability. However, we can resonate cloud margins with the insights we have on existing cloud businesses, e.g. from cloud providers. The sensitivity analysis shown in Figure 18 highlights what we see as key drivers for financial performance and investor assessment via valuation. Variations in EBITDA margins exert a slightly lower influence on long-term outcomes, whereas changes in revenue growth assumptions more materially affect projected profitability as measured in ROE. This is also reasonable as transparency on data center profitability can already be better judged from existing operations than the unknown revenue potential from additional AI adoption and speed.

Figure 17: EBITDA margin development and peer range (2016–2030E)



Sources: Bloomberg Intelligence, Allianz Research

With the profitability engine under scrutiny, ROE sensitivity will gain attention on top of sales growth. Within our financial analysis, capex monetization via revenue growth therefore emerges as the primary focus channel for financial performance, which we gauge by ROE. At the same time, the analysis highlights a delicate balance between investment intensity and returns. Our capex-to-sales sensitivity further illustrates this trade-off, as higher investment intensity would reduce ROE across identical growth scenarios, with returns declining from around c.25% to c.23–24%, even before considering potential margin pressure from faster hardware depreciation. In this framework, a capex-to-revenue ratio of around 23%, as discussed in the previous analysis of AI investment dynamics, would broadly correspond to the +5% capex scenario in our sensitivity table and therefore reflects a moderate 0.4pp increase in investment intensity relative to the base case. At the same time, the previously discussed estimate of around 200bps margin compression from faster hardware depreciation would broadly be captured by the -2% EBIT margin scenario in the sensitivity analysis and would translate into roughly a 1pp reduction in ROE, broadly consistent with the sensitivity illustrated in our framework. This indicates that expanding the capital base alone does not enhance profitability unless supported by sufficient revenue monetization. Big Tech companies rush into capex to ensure market share in the structural growth trend. This comes at the risk of higher capital base with insufficient monetization. A lower capital expenditure ratio combined with stronger revenue realization support higher ROE outcomes. Big Tech companies therefore appear to operate along a narrow path between sustained investment and maintaining stable return expectations.

Table 4: ROE sensitivity to growth, margin and capex assumptions

ROE		EBIT-Margin Increase				
		-5%	-2%	0%	2%	5%
5Y Sales CAGR	-5%	21.81%	22.59%	23.11%	23.63%	24.41%
	-2%	22.10%	22.89%	23.42%	23.94%	24.73%
	0%	22.29%	23.09%	23.62%	24.15%	24.94%
	2%	22.49%	23.29%	23.82%	24.36%	25.16%
	5%	22.78%	23.59%	24.13%	24.67%	25.48%

ROE		5Y Sales CAGR				
		-10%	-5%	0%	5%	10%
CAPEX	-10%	24.35%	24.86%	25.36%	25.87%	26.37%
	-5%	23.92%	24.43%	24.93%	25.44%	25.94%
	0%	23.50%	24.00%	24.51%	25.01%	25.51%
	5%	23.07%	23.57%	24.08%	24.58%	25.09%
	10%	22.64%	23.14%	23.65%	24.15%	24.66%

Source: Allianz Research

ROE is key – beyond being a profitability indicator it is also one driver for fair valuation. Forward valuation metrics based on consensus estimates suggest broadly stable to slightly moderating P/E ratios from c. 35x this year to c. 23-24x in three years. Technology equities therefore remain premium-valued, yet forward multiples indicate limited scope for further expansion. Market expectations appear increasingly focused on realized earnings delivery rather than additional valuation rerating. In this environment, future equity performance is likely to depend more strongly on the successful monetization of AI-related investments and sustained top-line growth, while multiple expansion plays a diminishing role and rests on long-term growth. High ROE is a key determinant for investors to inform a "fair" valuation multiple because it directly drives future earnings growth from retained earnings. What actually matters is the spread between ROE and cost of equity. The latter is also influenced by AI reshaping companies' capex and sales via increased leverage and potentially higher revenue growth variability. Beta has increased from below 1.2 in 2016-24 to 1.5 in 2024-26. We apply our analysis on ROE to evaluate potential PE ranges for Big Tech and conclude that current valuations appear fair under current assumptions but with some sensitivity to sales and capex intensity. Via this multiplier, future financial performance will add an even higher impact for share price performance. The framework presented in Figure xx illustrates a sensitivity analysis, showing how different assumptions for five-year sales CAGR and capex intensity translate into varying long-term valuation outcomes, with the indicated PE centering around a c. 23x earnings multiple by 2030, consistent with a structurally strong but more mature profitability profile. Sensitivity on sales growth is highest and above implication from capex which anyhow is unlikely to drop notably soon while upside surprises are more likely and can be expected to dampen both, EPS and ROE via higher depreciation as well as PE.

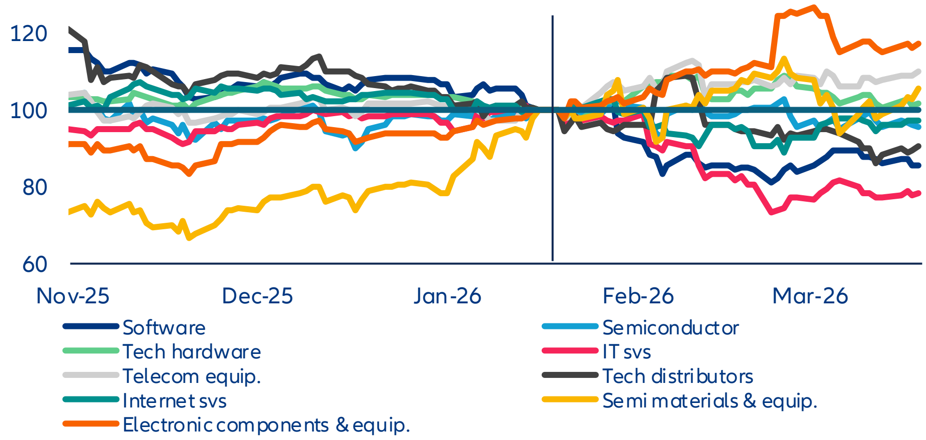
Table 5: Impact of growth and capital intensity on long-term valuation

P/E		Sales CAGR (surprise p.a.)						
		-9%	-6%	-3%	0%	3%	6%	9%
CAPEX (surprise p.a.)	-10%	23x	24x	25x	27x	28x	29x	31x
	-5%	22x	23x	24x	25x	26x	27x	29x
	0%	21x	21x	22x	23x	24x	25x	27x
	5%	20x	20x	21x	22x	23x	24x	25x
	10%	19x	19x	20x	21x	22x	22x	23x

Source: Allianz Research

The long-term outlook of the tech industry still looks rosy but uneven and inequal. We expect a more uneven performance trajectory for US technology stocks, with downside risks driven by valuation pressure and a broader shift toward risk-off positioning. Rotation away from megacap tech had already begun ahead of the Middle East escalation, suggesting that recent geopolitical events are amplifying, rather than initiating, the correction. Investor scrutiny is intensifying around the long-term monetization of elevated capex, particularly among hyperscalers, where the scale and timing of AI-related returns remain uncertain. At the same time, sentiment toward parts of the software sector is deteriorating, reflecting concerns over business model dilution as AI capabilities become more commoditized across corporate ecosystems. This combination is likely to sustain a degree of multiple compression and limit near-term upside for the most richly valued names. However, the outlook is not uniformly negative. Select segments within the tech value chain, notably semiconductors and infrastructure suppliers, continue to benefit from strong datacenter demand, with pricing dynamics potentially reinforced by rising commodity costs linked to geopolitical tensions. These areas appear structurally better positioned to capture near-term earnings visibility. In parallel, US leadership in AI infrastructure could attract incremental capital reallocation, particularly from Middle Eastern investors whose domestic investment pipelines may face delays. Over the longer term, the structural case for US tech remains intact, supported by innovation leadership and sustained demand for AI capabilities. That said, performance is likely to become more selective, with greater dispersion across subsectors and a less uniformly supportive environment than in previous cycles.

Figure 18: US equity performance of technology subsectors (rebased to 100 on 19 January)



Data as of 19 March 2026. Sources: LSEG Datastream, Allianz Research

These assessments are, as always, subject to the disclaimer provided below.

Allianz Research | 23 March 2026

Signal without response: Why the EU ETS needs resolve, not redesign

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In Summary

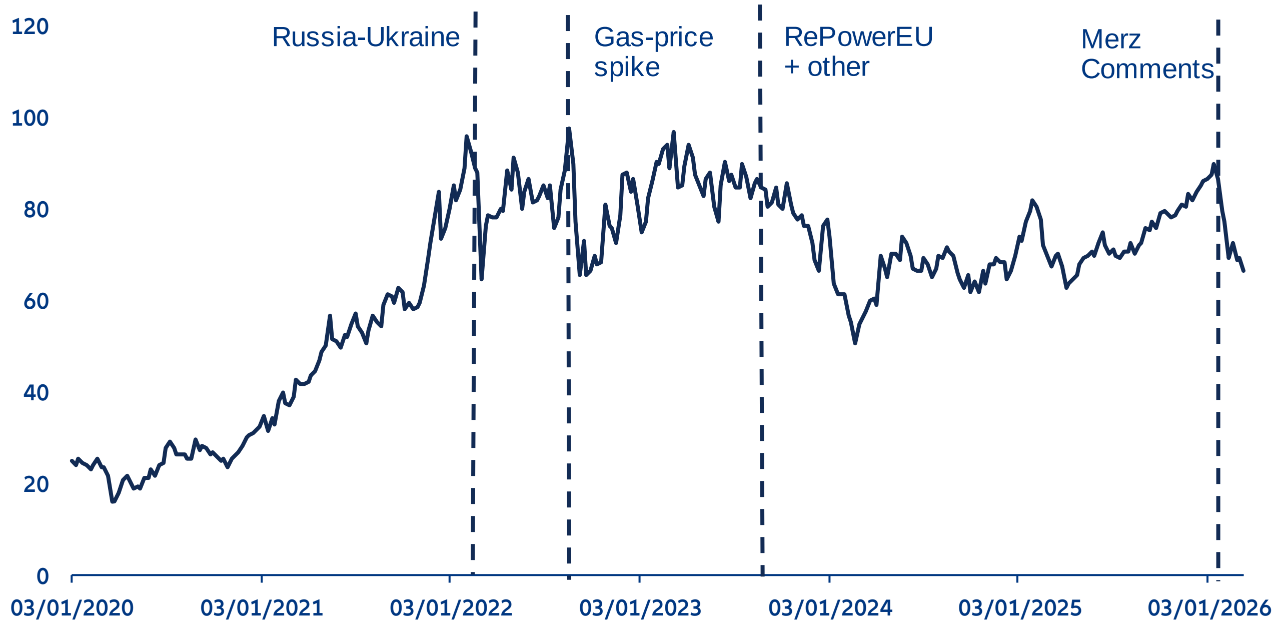
- **The EU Emissions Trading System (ETS) is at a political and structural inflection point, with allowance prices falling sharply by around -24% in the first two months of 2026. The conflict in Iran and associated energy price and inflation expectations have additionally spurred proposals to use the ETS as a lever for short-term price relief.** Unlike earlier episodes of price volatility rooted in energy-market fundamentals, the current turbulence reflects growing political questioning of the ETS's long-term design, compounded by concerns over industrial competitiveness as free-allowance phase-outs begin. Geopolitical pressure around Iran adds to this risk. The use of ETS revenues for price support or direct market intervention risk defeating the steering purpose of the system or creating dangerous lock-in effects.
- **Three ETS fault lines: free allocation has shielded more than 90% of industrial emissions from direct carbon costs but this is set to change; the ETS has driven substantial emission reductions in the power sector but not in industry and revenue recycling is falling short of its potential.** At current prices, full exposure would cost 0.9% of gross value added in the industry sector, a real burden for energy-intensive firms that cannot pass carbon costs through to customers given high energy costs, strong international competition and the limited offsetting potential of the Carbon Border Adjustment Mechanism (CBAM). Since 2005, power sector emissions have fallen by -54%, while industrial combustion emissions declined by a more moderate -33%. Renewables have made decarbonization viable at prices below EUR50/tCO2, while industrial options such as green hydrogen and low-carbon steel carry abatement costs well in excess of EUR100/tCO2. Persistent price uncertainty, with allowances oscillating in a EUR50–98/tCO2 corridor since 2022, has further weakened the investment case for capital-intensive industrial transformation. Last, only 16% of recycled revenues flow back into energy and industry, the sectors directly covered by the ETS, while 51% is directed toward non-covered sectors. A cumulative EUR41bn went undeployed for climate action between 2013 and 2024, risking a financing trap where industry bears rising carbon costs without the capital needed to decarbonize.
- **ETS2 is better positioned than ETS1 but that advantage is eroding.** Unlike ETS1, where abatement economics were the binding constraint, low-carbon alternatives in transport and buildings are already cost-competitive. The remaining barriers are political and financial: EUR113bn in annual fossil-fuel subsidies undermining the price signal, policy reversals on fossil boiler phase-outs, unaffordable EVs and financing frameworks still unequipped to meet household-level transition costs at scale. With household exposure potentially reaching EUR420 per year by 2030, these are policy choices that should be addressed before the carbon price arrives in 2028.
- **The ETS does not need a redesign but rather targeted reforms to restore price credibility and close the investment gap.** Seven priorities stand out: establishing a credible long-run EUA price corridor; scaling up Carbon Contracts for Difference to bridge the abatement cost gap in hard-to-abate sectors; introducing binding revenue recycling standards linked to the new decarbonization bank; prioritizing shared infrastructure for CO_2 transport, hydrogen and industrial clusters; conditioning the free allowance phase-out on the availability of abatement technologies and transition funding; expanding CBAM to cover downstream sectors currently left exposed and ensuring ETS2 revenues are transparently earmarked for households most exposed to its regressive cost distribution.

The ETS under fire: Market volatility and political pushback

The EU Emissions Trading System (ETS) entered 2026 under unusual strain. After reaching a two-year high of over EUR92/tCO2 in January, the EU Allowance (EUA) price corrected sharply by -14% within the same month and another -10% in February, falling to approximately EUR69/tCO2 (Figure 1). While the initial drop was triggered by geopolitical uncertainty over US tariffs, the future of Greenland and Iran, the second leg of the decline came amid growing political critique from member states, including Germany and Italy, which called into question not only the phase-out schedule of free allowances, but the long-term design of the mechanism itself. This turbulence arrives at a particularly consequential moment as the European Commission conducts its review of the ETS to determine whether its architecture remains fit for purpose in light of evolving market conditions and the EU's 2040 climate targets.

The observed price volatility is not unprecedented within the ETS, but earlier episodes were driven by fundamentally different forces. Between 2020 and 2022, allowance prices rose sharply, reflecting the post-Covid demand recovery, strengthened EU decarbonization ambition and an accelerated free-allowance phase-out schedule. The Russian invasion of Ukraine in February 2022 reversed this trajectory as anticipated gas supply disruptions curtailed industrial output and reduced compliance demand. Prices weakened further in 2023 amid declining industrial activity and the injection of additional allowances to finance RePowerEU. As Europe's energy dependence gradually shifted from Russian pipeline gas to US LNG, geopolitical risk remained the principal external driver of price formation, with US trade policy increasingly substituting pipeline politics as the key source of market uncertainty. Across these episodes, price movements were primarily rooted in energy market fundamentals or deliberate supply-side interventions, rather than in a broader questioning of the ETS's institutional design.

Figure 1: ETS allowance price and key market drivers (EUR/tCO2)



Sources: LSEG, Allianz Research

The key distinction from earlier episodes lies in the structural changes to the ETS taking effect from 2026, compounded by broader concerns about industrial electricity prices. From 2026 onwards, the free allowances allocated to European industry under the ETS begin to be phased out. The rationale was straightforward: as the Carbon Border Adjustment Mechanism (CBAM) enters into force to address international competitiveness concerns, the justification for shielding domestic industry through free allocation would progressively diminish, allowing the carbon price signal to drive further decarbonization. In practice, however, the transition has proved more contentious than anticipated. The CBAM, as finally adopted, covers a narrower set of sectors and a more limited range of emissions than originally proposed, leaving significant parts of European industry, particularly in downstream sectors, exposed to carbon costs without equivalent border protection. This concern is further reinforced by electricity prices that, while lower than their post-energy crisis peaks, remain structurally higher than those faced by key international competitors. At an EUA price of EUR65/tCO2, the carbon cost component of industrial electricity bills can reach up to 9% in the most carbon-intensive power systems, which disproportionately affects energy-intensive firms for which energy costs represent a larger fraction of total expenditure.[1]

Against this backdrop, political reactions have been swift. In Germany, the government announced plans to introduce a subsidized industrial electricity price of EUR0.05/kWh from 2026, roughly a third of the prevailing market rate, applicable to approximately 2,000 energy-intensive companies at an estimated fiscal cost of EUR3-5bn over three years. In Italy, a 2026 energy decree worth over EUR3bn proposed reimbursing gas-fired power plants for their ETS compliance costs, a measure that, given the role of gas in setting marginal electricity prices, would effectively decouple wholesale power prices from the carbon price signal. The European Commission has since opened a formal review of the measure's compatibility with EU state aid rules. Taken together, these interventions reflect a broader pattern in which the convergence of elevated energy costs and rising direct carbon-cost exposure is prompting member states to pursue unilateral measures that risk fragmenting the role of the ETS as a pan-European pricing instrument.

The conflict in Iran and associated energy price and inflation expectations have additionally spurred proposals to use the ETS as a lever for short-term price relief. These range from deploying ETS revenues for energy price support and adjusting the free allowance removal schedule to calls for direct market intervention to curb volatility. While some of these measures, such as the proposed ETS financed EUR30bn decarbonization booster, have their merits, others, such as the use of ETS revenues for price support or direct market intervention, risk defeating the steering purpose of the system or creating dangerous lock-in effects. The review of the ETS which was advanced to July 2026 is important and adjustments to the system can be warranted, however, the mechanism should not be misused as a short-term energy price management tool. For this purpose, other measures such as reducing non-steering levies on electricity including VAT, temporary state aid mechanisms for the most exposed energy-intensive industries, or targeted household transfers have a more direct and targeted impact on energy costs without undermining the carbon price signal the ETS is designed for.

Carbon costs in context: Limited exposure, rising trajectory

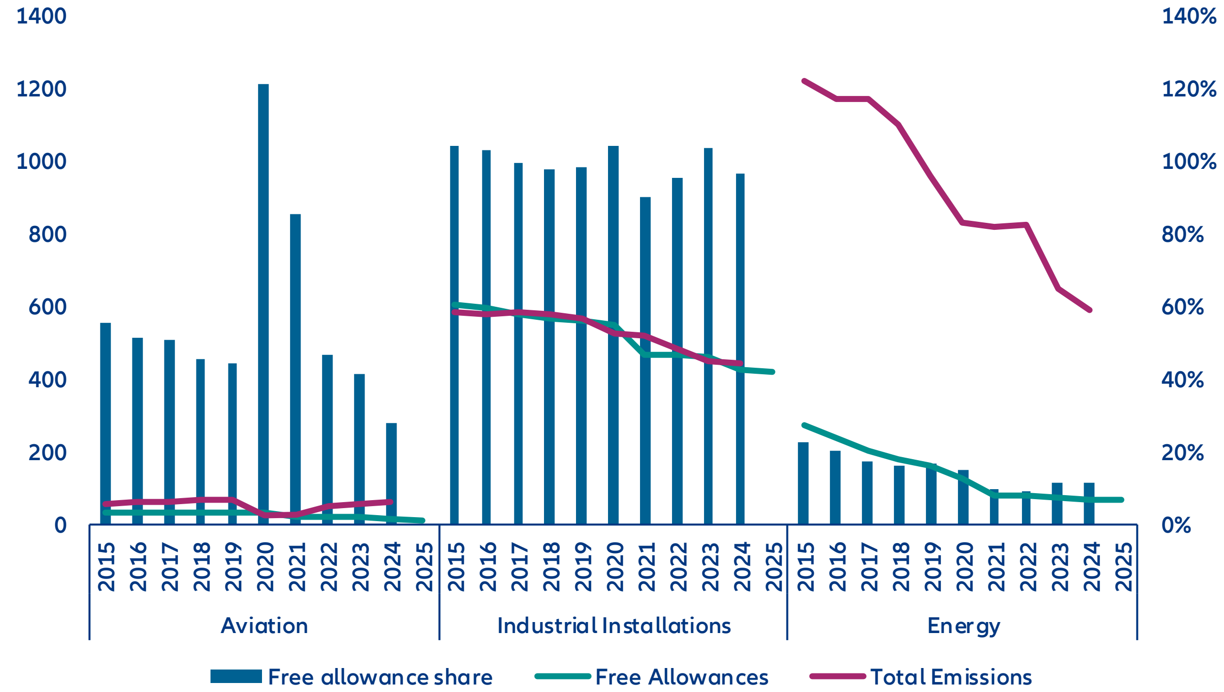
Despite the political concerns, the direct cost burden imposed by the ETS on European industry has thus far been limited. As shown in Figure 2, more than 90% of Scope 1 emissions in covered industrial sectors have been offset by free allowance allocation, effectively shielding most of the sector from direct carbon-cost exposure while some companies even generated profits from selling surplus allowances on the market. Even in a theoretical scenario with full emission coverage, ETS compliance costs in industry would amount to approximately EUR29bn , or around 0.9% of gross value added (GVA) across European industry in 2024 (Figure 3). The power sector bears a considerably higher burden in relative terms, at EUR38bn or 11.3% of sectoral GVA.

The concern is therefore less about the current level of exposure than its trajectory. Unlike utilities, industrial companies have limited ability to pass higher carbon costs through to consumers, making them more directly exposed to margin compression as free allowances are phased out. This is compounded by the gradual reduction in the overall supply of allowances, which will place upward pressure on EUA prices over time even

[1] ECB

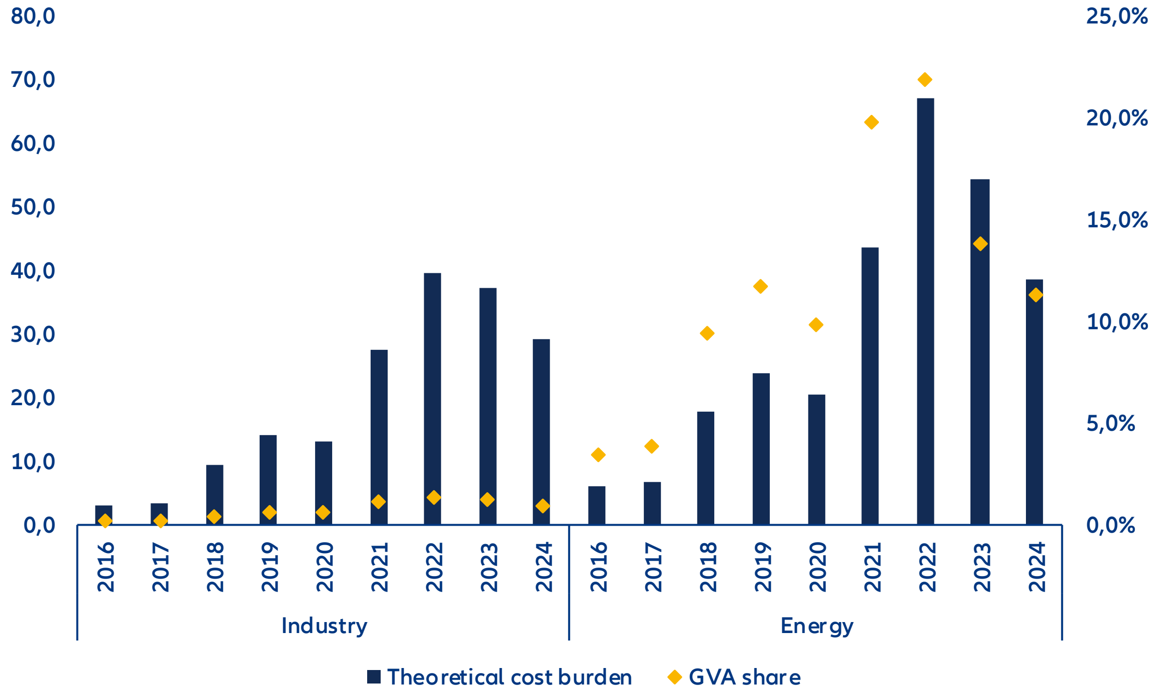
absent further policy changes. While the aggregate burden appears manageable at the economy-wide level, it is distributed unevenly across sectors and firm types, with energy-intensive industries facing the sharpest increase in net carbon costs as their allocation diminishes. A further structural issue is that the tradability of free allowances has not, in practice, generated the expected pace of industrial decarbonization. While the opportunity cost of holding rather than selling surplus allowances should in theory incentivize emission reductions, this mechanism has not translated into meaningful abatement investment across much of European industry. Understanding why requires looking beyond cost exposure to the sector-specific barriers that the carbon price signal has so far failed to overcome.

Figure 2: Sector emissions and free allowances (MtCo2, lhs) and free allowance share in total emissions (%, rhs)



Sources: EEA, Allianz Research

Figure 3: Theoretical cost burden under full exposure (EUR bn, lhs) and share in sector gross value-added (%, rhs)

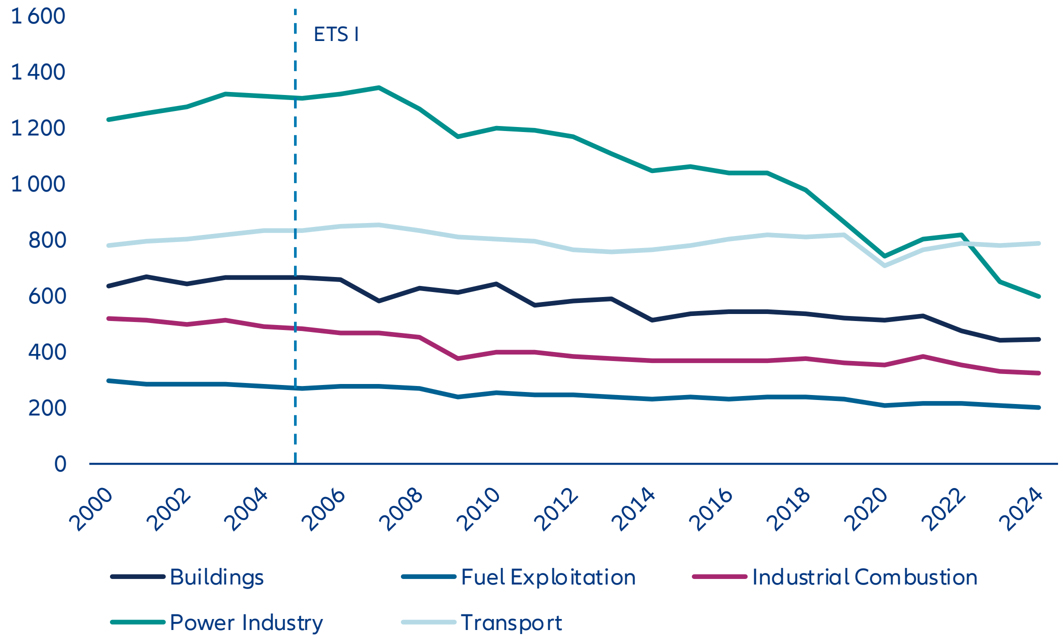


Sources: LSEG, Allianz Research

A tale of two sectors: Why carbon pricing works better for power than industry

The EU ETS has proven effective as a carbon-pricing instrument, yet its decarbonization impact varies markedly across sectors. Since its inception in 2005, emissions in covered sectors have fallen by more than -48%, compared with only -19% in non-covered sectors (Figure 4). However, the reduction is largely driven by the power sector, where emissions declined by -54%, while European industrial combustion emissions experienced a meaningful but more moderate decline of around -33%, reaching 324 MtCO2 in 2024. While free allocation is one mitigating factor in industrial decarbonization, firms retain incentives to abate as surplus allowances can be sold on carbon markets. The more fundamental constraints lie in sector-specific abatement economics. The power sector benefits from relatively short investment cycles, readily available low-carbon substitutes and competitive market structures that facilitate fuel switching. Industry, by contrast, faces longer capital stock turnover, technologically embedded process emissions and decarbonization options with higher marginal abatement costs – that is, higher costs for reducing an additional ton of CO2. These challenges are compounded by stronger exposure to international competition, which limits firms' ability to pass carbon costs through to product prices. As a result, carbon pricing may exert weaker investment and abatement signals in trade-exposed industrial sectors, even when allowances remain tradable.

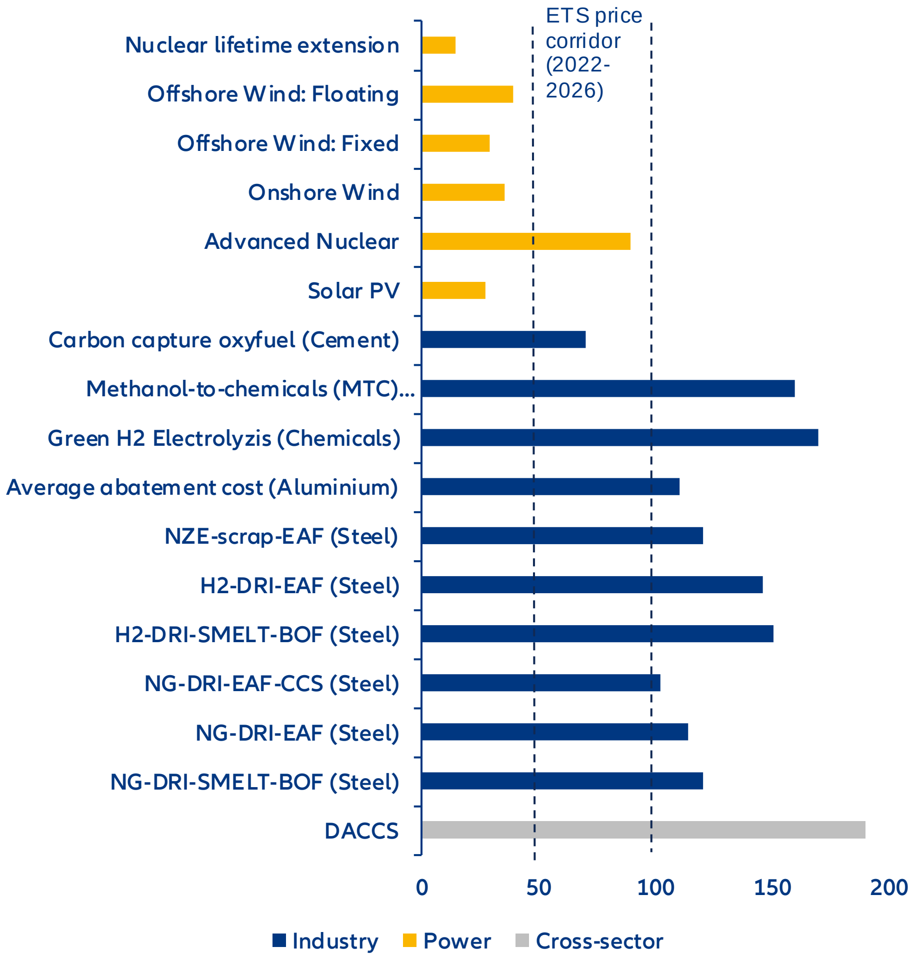
Figure 4: EU 27 emissions trajectory since introduction of EU ETS (MtCO2eq)



Sources: JRC EDGAR

High marginal abatement costs and price uncertainty represent the two principal barriers to industrial decarbonization under the current ETS framework. While the power sector has benefited from dramatic cost reductions in renewables that enable technology switching at prices below EUR50/tCO2, industrial decarbonization options such as green hydrogen, low-carbon steel and carbon capture typically carry abatement costs well in excess of EUR100/tCO2 (Figure 5). This cost gap reflects both the technological complexity of transforming energy-intensive production processes and the fact that many promising abatement technologies have yet to reach commercial maturity, leaving firms exposed to considerable cost and performance risk. Persistent price uncertainty adds a further barrier: although the post-Covid tightening of allowance supply did generate upward price pressure, since the energy crisis of 2022 allowance prices have largely oscillated within a EUR50–98/tCO2 corridor with considerable volatility. For capital-intensive investments with long payback periods and large upfront commitments, this unpredictability weakens the investment case precisely where long-term price credibility is most needed, and risks deferring the structural transitions that deep industrial decarbonization requires.

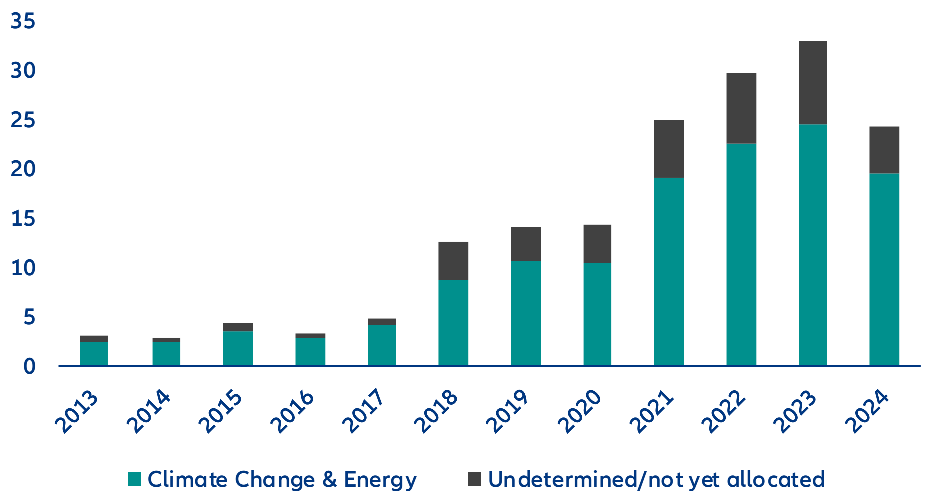
Figure 5: Abatement costs of selected decarbonization technologies by sector (EUR/tCO2)



Source: Allianz Research based on Agora & Wuppertal Institute [1, 2], MPP, EDF and IEA; Note: Abatement cost estimates can vary widely depending on sector, technology and region specific factors. Depicted values indicate cost averages.

ETS revenue recycling represents a significant underutilized lever for bridging the industrial-abatement cost gap. While the ETS obliges member states to reinvest allocated revenues in climate-related action, the mechanism underperforms in two key respects: the completeness of fund deployment and the strategic targeting of investments. Between 2013 and 2024, 13–31% of ETS revenues per year have remained unallocated or lacked a specific climate purpose, producing a cumulative deficit of approximately EUR41bn in effective revenue recycling (Figure 6). This structural shortfall limits the extent to which the ETS revenue mechanism can complement carbon price signals and support industrial transformation.

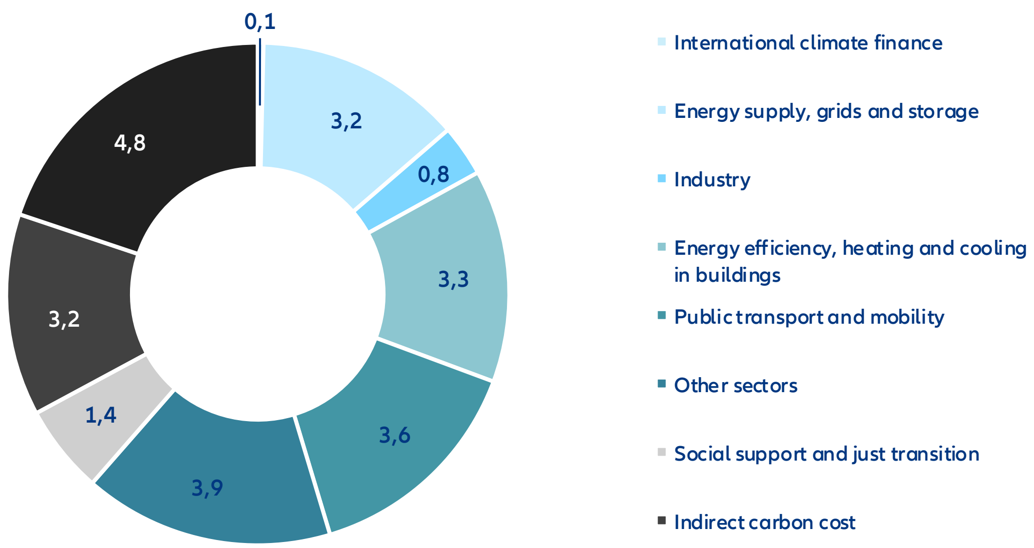
Figure 6: ETS revenue allocation gap (EUR bn)



Sources: EEA, Allianz Research

Of the funds that are recycled, the sectoral distribution raises further concerns. Only around 16% flows back into energy and industry transformation, the sectors directly covered under the ETS, while 44% is directed toward non-covered sectors such as buildings and transport (Figure 7). Although such cross-subsidization can be justified as a means of capturing near-term abatement opportunities and preparing these sectors for ETSII, an insufficient share returning to covered industries risks creating a financing trap, whereby firms bear the cost of carbon pricing without adequate access to the capital required to decarbonize. Equally important is the nature of recycling itself: in 2024, 13% of ETS revenues were directed toward "indirect carbon cost compensation" for energy-intensive industries, offsetting the very cost pressure the ETS is designed to create and thereby weakening the incentive to decarbonize. The same is true of support directed at technologies that have already reached cost-competitiveness, such as feed-in tariff schemes for mature renewables, where additionality is limited. To improve the efficiency of ETS revenue recycling, all revenues should be directed toward high-additionality decarbonization activities, ensuring covered sectors are not left underfinanced in the transition.

Figure 7: ETS revenue allocation by sector and purpose (%)

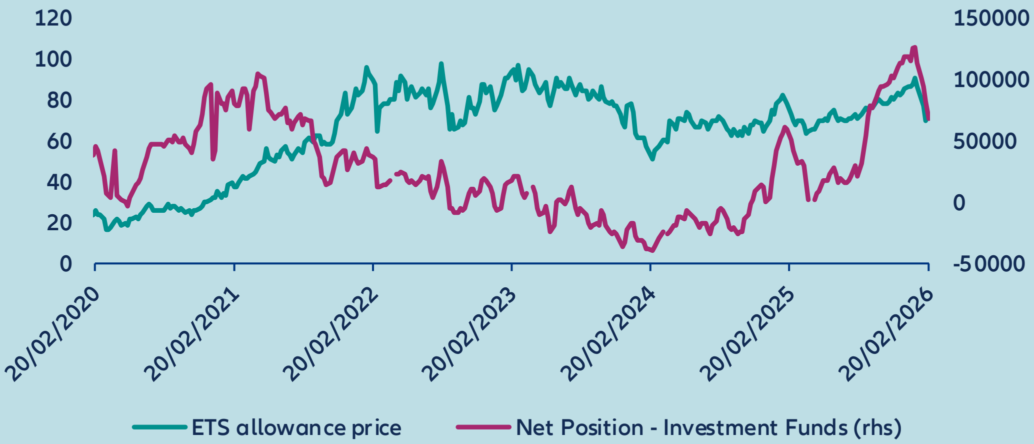


Sources: LSEG, Allianz Research

BOX 1: What is the effect on capital markets?

The financial sector constitutes a primary participant in EU ETS secondary-market trading by volume. While non-financial corporates and compliance entities accumulate net long positions, banks predominantly serve as counterparties on the short side. Investment funds represent a smaller but notable participant cohort, exhibiting positioning dynamics that more closely reflect prevailing market expectations for price developments. Following the Russian invasion of Ukraine, funds rotated to net short positions, consistent with anticipated contractions in industrial output and EUA demand (Figure 8). This positioning reversed to net long in November 2024, before funds began unwinding exposure in early January 2025, contributing to the pronounced price decline observed over that period. Prior to the price decline, investment funds had built up a record net-long position.

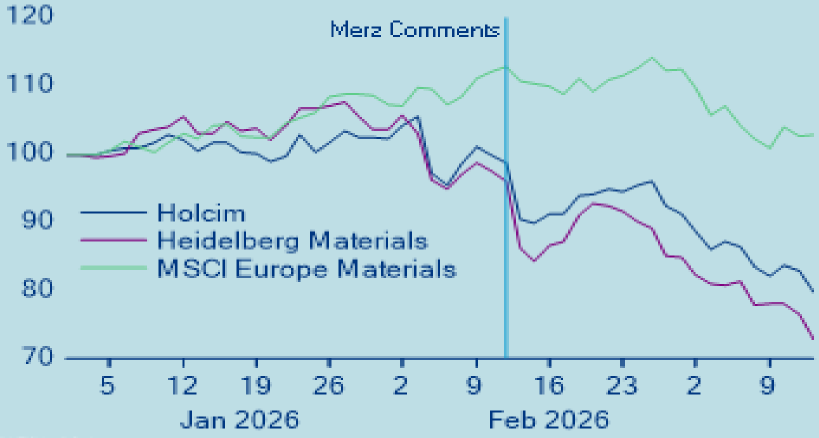
Figure 8: Investment funds' net positioning in EUA futures



Sources: Bloomberg, Allianz Research

The ongoing policy debate surrounding the future of the EU ETS has materially influenced investor expectations, particularly within the cement sector. Following the news of discussions on the future of the EU ETS market, the share price of Heidelberg Materials and Holcim, which are seen as transition leaders in the industry (see Transition Pathway Initiative assessment, for example), underperformed the broader materials sector (Figure 9). A prolonged phase-out of free allowances would disproportionately disadvantage companies that have already committed significant capital to decarbonization. Firms that invested early in low-carbon technologies or operational upgrades incurred substantial costs in anticipation of a tightening regulatory environment and rising carbon prices – an environment in which their abatement investments would generate a clear competitive return. An extension of free allocation effectively subsidizes higher-emitting peers, compressing the cost differential between decarbonization leaders and laggards and eroding the competitive advantage that early movers sought to secure. Simultaneously, lower prevailing CO_2 prices reduce the financial viability of capital-intensive abatement technologies such as Carbon Capture, Utilization and Storage (CCUS), for which the economic case depends critically on a sufficiently high and predictable carbon price signal. In aggregate, policy signals pointing toward a more lenient allocation regime risk undermining both the relative positioning of decarbonization leaders and the broader investment case for emission-reduction technologies across the sector.

Figure 9: Price reactions in European cement equities following statements on EU ETS reform



Sources: LSEG, Allianz Research

EUA price dynamics represent a meaningful driver of equity returns in carbon-intensive sectors. To quantify this relationship, sector index returns were regressed on both STOXX Europe 600 returns and the returns of rolled December EUA futures contracts from 2022 onwards. The results reveal a statistically significant and negative coefficient for EUA returns in the case of the STOXX Europe 600 Construction Materials Index and the STOXX Europe 600 Chemicals Index, indicating that rising carbon prices weigh on equity performance in these sectors. By contrast, no such relationship was identified for the STOXX Europe 600 Utilities or Basic Resources subindices. In the case of Basic Resources, this outcome is largely a function of index composition, given the limited share of steel producers among its constituents. For Utilities, the absence of a significant carbon price sensitivity is more structural in nature, reflecting the sector's substantial progress in decarbonization over recent years and the correspondingly reduced reliance on purchased allowances.

For now the market seems to price in an extreme scenario of adjustments to the EU ETS for some of the affected companies rather than a delayed phase out. Changes to the EU ETS are necessary to improve its effectiveness, but the phase-out of free allowances should continue. For investors aiming to decarbonize their portfolios and invest into transition leaders, these market reactions present a good opportunity to build up exposure to companies in hard-to-abate sectors that are actively decarbonizing their operations and that have credible transition plans.

Bracing for impact: Getting buildings and transport ready for ETS2

The EU ETS experience offers both a proof of concept and a cautionary tale for the next phase of European carbon pricing. Confirmed to begin in 2028 following a one-year delay, ETS2 will extend carbon pricing to fuel combustion in buildings and road transport, sectors that together account for 52% of EU greenhouse gas emissions and where decarbonization progress has significantly lagged. Unlike ETS1, ETS2 will operate as a separate emissions trading system with its own distinct carbon price, reflecting the different cost structures and abatement dynamics of these sectors. Regulated entities will be fuel distributors rather than end-users, who will bear the cost indirectly through fuel prices. To limit affordability pressures during the start-up phase, ETS2 includes a transitional price cap of EUR 45/tCO2 in 2023 real terms (approximately EUR59/tCO2 in nominal 2028 terms), below which the carbon price is prevented from rising freely. The central lesson from ETS1 is that carbon pricing works when abatement alternatives are cost-competitive and enabling infrastructure is in place before the price signal bites. The cap is therefore an acknowledgment that those conditions are not yet fully met, and that the transition needs time to mature before the full price signal can do its work.

Cost-competitive low-carbon alternatives exist in both sectors, but remain underdeployed. In road transport, battery costs have fallen by over -75% since 2015, yet European consumers have not fully benefited. Pack prices in Europe remain around 56% above Chinese levels, and carmakers' focus on high-margin premium models pushed the average EU EV price up by EUR5,000 between 2020 and 2024 to EUR44,600.[2] Passing battery cost reductions through to vehicle prices, alongside a continued expansion of charging infrastructure, will be essential to make EVs cost-competitive before ETS2 enters into force. In buildings, accelerating energy-efficiency renovations, expanding heating networks in urban areas and scaling heat pump adoption are the central levers, yet all three face high upfront costs that current financing frameworks are not yet equipped to meet at scale.[3]

Closing these gaps before 2028 is urgent as inaction carries a measurable cost. In a pessimistic scenario with full sector coverage and only marginal reductions in emissions over the coming years, the annual ETS2 cost could be as high as EUR69bn by 2028 and EUR114bn by 2030 (Figure 10). Should ETS2 achieve its targeted 42% emission reduction against 2005 levels, these figures fall to EUR58bn in 2028 and EUR85bn in 2030. While this would alleviate some of the pressure on households, decreasing the cost burden by 15%-26%, the annual additional household expenses could come in as high as EUR420 per year by 2030. Reducing these costs is achievable, but hinges on adequate redistribution and well-designed supporting policies.

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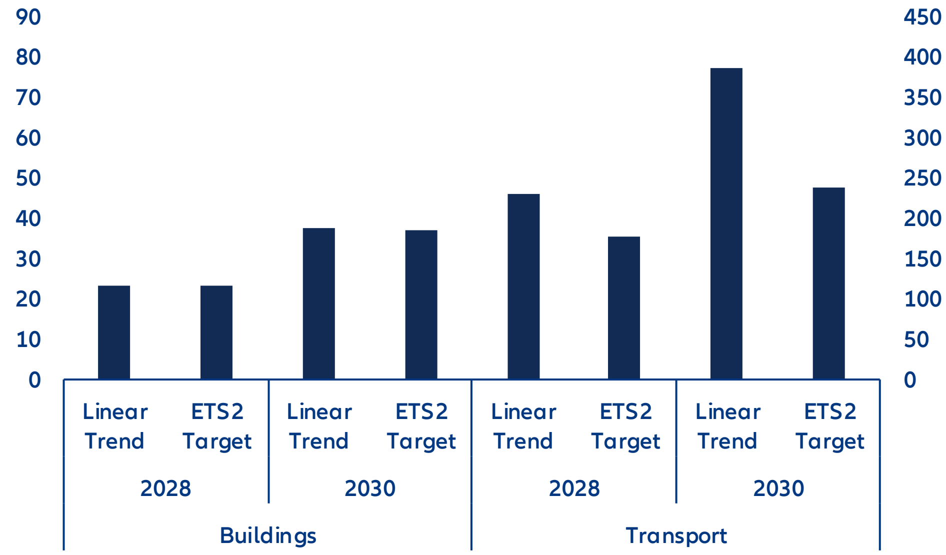
[2] *BNEF and T&E*

[3] *For a more detailed analysis on heat pumps see The heat is on: Unlocking Germany's heat-pump potential*

Figure 10: Projected ETS2 revenues per sector: annual total (EUR bn; lhs) and per household (EUR; rhs) under full sectoral coverage



Sources: JRC EDGAR, Eurostat, Allianz Research; Note: For 2028 an initial maximum allowance price of EUR59/tCO2 was assumed which increases to EUR100/tCO2 in 2030; The ETS Target scenario is based on the 42% ETS reduction target for 2030 while the linear trend assumes an annual emission reduction rate aligned with the average annual reduction over the last 20 years

Making ETS2 work requires addressing these conditions well before the carbon price arrives. ETS1 revenues have already been partially directed toward buildings and transport, providing a foundation on which ETS2 can build, but the gaps remain substantial. Fossil-fuel subsidies across the EU still amount to EUR113bn annually, suppressing the relative cost of carbon-intensive heating and driving and blunting the investment case for cleaner alternatives. A credible partial phase-out would simultaneously strengthen price incentives and free up fiscal space for the infrastructure and household support the transition requires. Equally important is policy consistency as repeated reversals on fossil boiler phase-outs, subsidy schemes and regulatory timelines have fostered a wait-and-see mentality among households and investors, deferring the long-cycle decisions on heat pump installations, building retrofits and EV purchases that need to be made within the coming years.

Social acceptability is the defining political challenge for ETS2, given that its costs fall more directly on households. The aggregate macroeconomic impact is manageable: at a carbon price of EUR 59/tCO2, the estimated inflationary effect amounts to approximately 0.54pp, while the GDP impact is projected at around 0.02pp provided revenues are recycled rather than absorbed into general fiscal accounts.[4] However, these aggregate figures mask a regressive distribution, with lower-income households facing a disproportionate burden as energy and transport represent a larger share of their expenditure. Ensuring that revenues are recycled toward those most exposed, through direct transfers, subsidized access to low-carbon technologies and investment in shared infrastructure, will therefore be central to maintaining the public legitimacy on which ETS2's long-term viability depends.

From price signal to investment signal: Seven priorities for ETS reform

The EU ETS is fundamentally sound, but the political commitment to making it work has not matched its ambition; what the system requires is not a redesign but the resolve to make it effective and predictable. The core challenge is not that the ETS is too stringent, but that it has failed to provide the certainty and complementary support that capital-intensive industries require to commit to deep decarbonization. That failure carries a forward-looking dimension: With ETS2 set to extend carbon pricing to buildings and road transport from 2028, the structural lessons of ETS1 – on price credibility, revenue recycling and enabling infrastructure –

[4] *See Climate policy: Europe's industrial shield or a drag on competitiveness?*

are also the design requirements that will determine whether the next phase succeeds. The following seven recommendations address this commitment gap and represent the priorities the Commission should act on in its 2026 ETS review:

1. **A credible and predictable EUA price path is a prerequisite for long-term investment planning in carbon-intensive industries.** Many of the most consequential decarbonization decisions, such as switching to hydrogen as an industrial feedstock, involve long capital cycles and require sufficient and stable carbon price expectations to be financially justified. The Market Stability Reserve has broadly fulfilled its role in correcting surplus-driven price weakness, but should be refined to better contain sharp volatility and support a credible long-run price corridor. Publishing a transparent indicative price trajectory would materially reduce investment uncertainty and strengthen the carbon price as a basis for industrial decision-making.
2. **Bridging the abatement cost gap requires scaling up Carbon Contracts for Difference (CCfDs) as a primary instrument for de-risking investment in breakthrough technologies.** CCfDs work by guaranteeing producers a fixed return for low-carbon production, paying out the difference between an agreed strike price and the prevailing carbon price when the latter falls short of what is needed to make the investment viable. This ensures that decarbonization investments in hard-to-abate sectors such as steel, cement, and chemicals remain financially attractive independent of carbon price volatility. Scaling up CCfD programmes and ensuring sufficient public backing to underwrite strike price commitments should therefore be a central priority of the ETS reform agenda.
3. **Revenue recycling must be made more effective through binding standards that prevent underspending and ensure funds are directed toward activities where additional decarbonization can be achieved.** The newly established decarbonization bank provides a potential vehicle for channeling industrial transformation funding at scale, and its mandate should be explicitly linked to ETS revenue flows. Alongside this, a coordination mechanism is needed to prevent unilateral national measures, such as subsidized industrial electricity prices or ETS cost reimbursements, from fragmenting the integrity of the pan-European carbon price signal. The same discipline must apply to ETS2 revenues from the outset: the EUR113bn in annual fossil-fuel subsidies across the EU that currently suppress the relative cost of carbon-intensive heating and transport represent both a fiscal and a political opportunity. A credible phase-out would simultaneously strengthen price incentives and free up resources for the household support and infrastructure investment that ETS2 requires.
4. **Decarbonization of energy-intensive industries also depends on shared infrastructure that no individual firm can finance alone.** Public investment should prioritize CO2 transport and storage networks, hydrogen pipelines and the development of industrial clusters where infrastructure costs can be pooled. Without this foundation, even well-funded firms face structural barriers to acting on the carbon price signal.
5. **The phase-out of free allowances should continue, but its pace must be made more explicitly contingent on the availability of abatement technologies, transition funding and infrastructure.** A phase-out that runs ahead of these enabling conditions risks creating a financing trap in which firms face rising compliance costs without a viable decarbonization pathway. The schedule should also incorporate a more systematic assessment of carbon leakage risk and competitive asymmetries, particularly for sectors not yet fully covered by CBAM.
6. **The Carbon Border Adjustment Mechanism should be expanded to cover downstream sectors and indirect emissions currently left exposed by its narrow adoption.** As free allocation is phased out, border protection must keep pace to ensure that European industry is not placed at a structural competitive disadvantage relative to international competitors operating under less stringent carbon regimes. A broader CBAM scope is therefore not only a competitiveness measure but a precondition for the phase-out of free allocation to proceed without undermining the industrial base the ETS is intended to transform.
7. **Social acceptability must be treated as a first-order design requirement for ETS2, not an afterthought.** Unlike ETS1, where carbon costs fall primarily on industrial firms, ETS2 will reach households directly through fuel prices for heating and road transport from 2028. The cost distribution

is regressive by nature – lower-income households spend a larger share of their income on energy and transport and will bear a disproportionate burden unless revenues are actively recycled toward direct transfers, subsidized access to low-carbon technologies and investment in shared infrastructure. Ensuring that ETS2 revenues are transparently earmarked for those most exposed is not merely an equity concern: it is the precondition for maintaining the public legitimacy on which the long-term viability of European carbon pricing depends.

These assessments are, as always, subject to the disclaimer provided below.